Linear Algebra Workbook Solutions

Orthogonality and change of basis

krista king
MATH

ORTHOGONAL COMPLEMENTS

1. Find the orthogonal complement of V, $V^{\perp}$.

$$V = \text{Span}\left(\begin{bmatrix} -2 \\ 1 \\ 4 \end{bmatrix}, \begin{bmatrix} 0 \\ -3 \\ 2 \end{bmatrix} \right)$$

Solution:

The subspace V is a plane in $\mathbb{R}^3$, spanned by $\vec{v}_1 = (-2,1,4)$ and $\vec{v}_2 = (0,-3,2)$. So its orthogonal complement $V^{\perp}$ is the set of vectors which are orthogonal to both $\vec{v}_1 = (-2,1,4)$ and $\vec{v}_2 = (0,-3,2)$.

$$V^{\perp} = \{ \vec{x} \in \mathbb{R}^3 \mid \vec{x} \cdot \begin{bmatrix} -2 \\ 1 \\ 4 \end{bmatrix} = 0 \quad \text{and} \quad \vec{x} \cdot \begin{bmatrix} 0 \\ -3 \\ 2 \end{bmatrix} = 0 \}$$

If we let $\vec{x} = (x_1, x_2, x_3)$, we get two equations from these dot products.

$$-2x_1 + x_2 + 4x_3 = 0$$

$$-3x_2 + 2x_3 = 0$$

Put these equations into an augmented matrix,

$$\left[\begin{array}{ccc|c} -2 & 1 & 4 & 0 \\ 0 & -3 & 2 & 0 \end{array}\right]$$

then put it into reduced row-echelon form.

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$$\begin{bmatrix} 1 & -\frac{1}{2} & -2 & | & 0 \\ 0 & -3 & 2 & | & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -\frac{1}{2} & -2 & | & 0 \\ 0 & 1 & -\frac{2}{3} & | & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -\frac{7}{3} & | & 0 \\ 0 & 1 & -\frac{2}{3} & | & 0 \end{bmatrix}$$

The rref form gives the system of equations

$$x_1 - \frac{7}{3}x_3 = 0$$

$$x_2 - \frac{2}{3}x_3 = 0$$

and we can solve the system for the pivot variables.

$$x_1 = \frac{7}{3}x_3$$

$$x_2 = \frac{2}{3}x_3$$

So we could also express the system as

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} \frac{7}{3} \\ \frac{2}{3} \\ 1 \end{bmatrix}$$

Which means the orthogonal complement $V^{\perp}$ is

$$V^{\perp} = \text{Span}\left(\begin{bmatrix} \frac{7}{3} \\ \frac{2}{3} \\ 1 \end{bmatrix} \right)$$

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2. Find the orthogonal complement of V, $V^{\perp}$.

$$V = \text{Span}\left(\begin{bmatrix} -1 \\ 2 \\ -5 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -4 \\ 3 \end{bmatrix} \right)$$

Solution:

The subspace V is a plane in $\mathbb{R}^4$, spanned by $\vec{v}_1 = (-1,2,-5,3)$ and $\vec{v}_2 = (1,0,-4,3)$. So its orthogonal complement $V^{\perp}$ is the set of vectors which are orthogonal to both $\vec{v}_1 = (-1,2,-5,3)$ and $\vec{v}_2 = (1,0,-4,3)$.

$$V^{\perp} = \{ \vec{x} \in \mathbb{R}^4 \mid \vec{x} \cdot \begin{bmatrix} -1 \\ 2 \\ -5 \\ 3 \end{bmatrix} = 0 \quad \text{and} \quad \vec{x} \cdot \begin{bmatrix} 1 \\ 0 \\ -4 \\ 3 \end{bmatrix} = 0 \}$$

If we let $\vec{x} = (x_1, x_2, x_3, x_4)$, we get two equations from these dot products.

$$-x_1 + 2x_2 - 5x_3 + 3x_4 = 0$$

$$x_1 - 4x_3 + 3x_4 = 0$$

Put these equations into an augmented matrix,

$$\left[\begin{array}{cccc|c} -1 & 2 & -5 & 3 & 0 \\ 1 & 0 & -4 & 3 & 0 \end{array}\right]$$

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then put it into reduced row-echelon form.

$$\left[\begin{array}{cccc|c} 1 & -2 & 5 & -3 & 0 \\ 1 & 0 & -4 & 3 & 0 \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & -2 & 5 & -3 & 0 \\ 0 & 2 & -9 & 6 & 0 \end{array}\right]$$

$$\left[\begin{array}{cccc|c} 1 & -2 & 5 & -3 & 0 \\ 0 & 1 & -\frac{9}{2} & 3 & 0 \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & 0 & -4 & 3 & 0 \\ 0 & 1 & -\frac{9}{2} & 3 & 0 \end{array}\right]$$

The rref form gives the system of equations

$$x_1 - 4x_3 + 3x_4 = 0$$

$$x_2 - \frac{9}{2}x_3 + 3x_4 = 0$$

and we can solve the system for the pivot variables.

$$x_1 = 4x_3 - 3x_4$$

$$x_2 = \frac{9}{2}x_3 - 3x_4$$

So we could also express the system as

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = x_3 \begin{bmatrix} 4 \\ \frac{9}{2} \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -3 \\ -3 \\ 0 \\ 1 \end{bmatrix}$$

Which means the orthogonal complement $V^{\perp}$ is

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$$V^{\perp} = \text{Span}\left(\begin{bmatrix} 4 \\ \frac{9}{2} \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ -3 \\ 0 \\ 1 \end{bmatrix} \right)$$

■ 3. Rewrite the orthogonal complement of V, $V^{\perp}$, if V is a vector set in $\mathbb{R}^3$.

$$V = \begin{bmatrix} s \\ -2s - t \\ s + t \end{bmatrix}$$

Solution:

We can rewrite V as

$$V = \{ s \cdot \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} + t \cdot \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \mid s, t \in \mathbb{R}^3 \}$$

The subspace V is a plane in $\mathbb{R}^3$, spanned by $\vec{v}_1 = (1, -2, 1)$ and $\vec{v}_2 = (0, -1, 1)$. So its orthogonal complement $V^{\perp}$ is the set of vectors which are orthogonal to both $\vec{v}_1 = (1, -2, 1)$ and $\vec{v}_2 = (0, -1, 1)$.

$$V^{\perp} = \{ \vec{x} \in \mathbb{R}^3 \mid \vec{x} \cdot \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} = 0 \quad \text{and} \quad \vec{x} \cdot \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} = 0 \}$$

If we let $\vec{x} = (x_1, x_2, x_3)$, we get two equations from these dot products.

$$x_1 - 2x_2 + x_3 = 0$$

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$$-x_2 + x_3 = 0$$

Put these equations into an augmented matrix,

$$\begin{bmatrix} 1 & -2 & 1 & | & 0 \\ 0 & -1 & 1 & | & 0 \end{bmatrix}$$

then put it into reduced row-echelon form.

$$\begin{bmatrix} 1 & -2 & 1 & | & 0 \\ 0 & 1 & -1 & | & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -1 & | & 0 \\ 0 & 1 & -1 & | & 0 \end{bmatrix}$$

The rref form gives the system of equations

$$x_1 - x_3 = 0$$

$$x_2 - x_3 = 0$$

and we can solve the system for the pivot variables.

$$x_1 = x_3$$

$$x_2 = x_3$$

So we could also express the system as

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Which means the orthogonal complement is

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$$V^{\perp} = \text{Span}\left(\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right)$$

■ 4. Rewrite the orthogonal complement of W, $W^{\perp}$, if W is a vector set in $\mathbb{R}^4$.

$$W = \begin{bmatrix} -2y - z \\ 3y + z \\ -y \\ 2y - 3z \end{bmatrix}$$

Solution:

We can rewrite W as

$$W = \{ y \cdot \begin{bmatrix} -2 \\ 3 \\ -1 \\ 2 \end{bmatrix} + z \cdot \begin{bmatrix} -1 \\ 1 \\ 0 \\ -3 \end{bmatrix} \mid y, z \in \mathbb{R}^4 \}$$

The subspace W is a plane in $\mathbb{R}^4$, spanned by $\vec{w}_1 = (-2,3,-1,2)$ and $\vec{w}_2 = (-1,1,0,-3)$. So its orthogonal complement $W^{\perp}$ is the set of vectors which are orthogonal to both $\vec{w}_1 = (-2,3,-1,2)$ and $\vec{w}_2 = (-1,1,0,-3)$.

$$W^{\perp} = \{ \vec{x} \in \mathbb{R}^4 \mid \vec{x} \cdot \begin{bmatrix} -2 \\ 3 \\ -1 \\ 2 \end{bmatrix} = 0 \quad \text{and} \quad \vec{x} \cdot \begin{bmatrix} -1 \\ 1 \\ 0 \\ -3 \end{bmatrix} = 0 \}$$

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If we let $\vec{x} = (x_1, x_2, x_3, x_4)$, we get two equations from these dot products.

$$-2x_1 + 3x_2 - x_3 + 2x_4 = 0$$

$$-x_1 + x_2 - 3x_4 = 0$$

Put these equations into an augmented matrix,

$$\left[\begin{array}{cccc|c} -2 & 3 & -1 & 2 & 0 \\ -1 & 1 & 0 & -3 & 0 \end{array}\right]$$

then put it into reduced row-echelon form.

$$\left[\begin{array}{cccc|c} 1 & -\frac{3}{2} & \frac{1}{2} & -1 & 0 \\ -1 & 1 & 0 & -3 & 0 \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & -\frac{3}{2} & \frac{1}{2} & -1 & 0 \\ 0 & -\frac{1}{2} & \frac{1}{2} & -4 & 0 \end{array}\right]$$

$$\left[\begin{array}{cccc|c} 1 & -\frac{3}{2} & \frac{1}{2} & -1 & 0 \\ 0 & 1 & -1 & 8 & 0 \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & 0 & -1 & 11 & 0 \\ 0 & 1 & -1 & 8 & 0 \end{array}\right]$$

The rref form gives the system of equations

$$x_1 - x_3 + 11x_4 = 0$$

$$x_2 - x_3 + 8x_4 = 0$$

and we can solve the system for the pivot variables.

$$x_1 = x_3 - 11x_4$$

$$x_2 = x_3 - 8x_4$$

So we could also express the system as

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$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = x_3 \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -11 \\ -8 \\ 0 \\ 1 \end{bmatrix}$$

Which means the orthogonal complement is

$$W^{\perp} = \text{Span}\left(\begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -11 \\ -8 \\ 0 \\ 1 \end{bmatrix} \right)$$

■ 5. Describe the orthogonal component of V, $V^{\perp}$.

$$V = \text{Span}\left(\begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 3 \\ 0 \end{bmatrix} \right)$$

Solution:

The subspace V is a plane in $\mathbb{R}^4$, spanned by the $\vec{v}_1 = (1, -1, 0, 1)$, $\vec{v}_2 = (-1, 2, 1, 3)$, and $\vec{v}_3 = (2, -1, 3, 0)$. So its orthogonal complement $V^{\perp}$ is the set of vectors which are orthogonal to $\vec{v}_1 = (1, -1, 0, 1)$, $\vec{v}_2 = (-1, 2, 1, 3)$, and $\vec{v}_3 = (2, -1, 3, 0)$.

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$$V^{\perp} = \{ \vec{x} \in \mathbb{R}^4 \mid \vec{x} \cdot \begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \end{bmatrix} = 0 \, , \vec{x} \cdot \begin{bmatrix} -1 \\ 2 \\ 1 \\ 3 \end{bmatrix} = 0 \quad \text{and} \quad \vec{x} \cdot \begin{bmatrix} 2 \\ -1 \\ 3 \\ 0 \end{bmatrix} = 0 \}$$

If we let $\vec{x} = (x_1, x_2, x_3, x_4)$, we get three equations from these dot products.

$$x_1 - x_2 + x_4 = 0$$

$$-x_1 + 2x_2 + x_3 + 3x_4 = 0$$

$$2x_1 - x_2 + 3x_3 = 0$$

Put these equations into an augmented matrix,

$$\left[\begin{array}{cccc|c} 1 & -1 & 0 & 1 & 0 \\ -1 & 2 & 1 & 3 & 0 \\ 2 & -1 & 3 & 0 & 0 \end{array}\right]$$

then put it into reduced row-echelon form.

$$\left[\begin{array}{cccc|c} 1 & -1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 4 & 0 \\ 2 & -1 & 3 & 0 & 0 \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & -1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 4 & 0 \\ 0 & 1 & 3 & -2 & 0 \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 1 & 5 & 0 \\ 0 & 1 & 1 & 4 & 0 \\ 0 & 1 & 3 & -2 & 0 \end{array}\right]$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 1 & 5 & 0 \\ 0 & 1 & 1 & 4 & 0 \\ 0 & 0 & 2 & -6 & 0 \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 1 & 5 & 0 \\ 0 & 1 & 1 & 4 & 0 \\ 0 & 0 & 1 & -3 & 0 \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 0 & 8 & 0 \\ 0 & 1 & 1 & 4 & 0 \\ 0 & 0 & 1 & -3 & 0 \end{array}\right]$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 8 & 0 \\ 0 & 1 & 0 & 7 & 0 \\ 0 & 0 & 1 & -3 & 0 \end{array}\right]$$

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The rref form gives the system of equations

$$x_1 + 8x_4 = 0$$

$$x_2 + 7x_4 = 0$$

$$x_3 - 3x_4 = 0$$

which we can solve for the pivot variables.

$$x_1 = -8x_4$$

$$x_2 = -7x_4$$

$$x_3 = 3x_4$$

So we could also express the system as

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = x_4 \begin{bmatrix} -8 \\ -7 \\ 3 \\ 1 \end{bmatrix}$$

Which means the orthogonal complement is

$$V^{\perp} = \text{Span}\left(\begin{bmatrix} -8 \\ -7 \\ 3 \\ 1 \end{bmatrix} \right)$$

6. Describe the orthogonal component of W, $W^{\perp}$.

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$$W = \text{Span}\left(\begin{bmatrix} 1 \\ 0 \\ 2 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} -3 \\ 2 \\ 4 \\ 1 \\ -2 \end{bmatrix} \right)$$

Solution:

The subspace W is a plane in $\mathbb{R}^5$, spanned by $\vec{w}_1 = (1,0,2,-1,2)$ and $\vec{w}_2 = (-3,2,4,1,-2)$. So its orthogonal complement $W^{\perp}$ is the set of vectors which are orthogonal to $\vec{w}_1 = (1,0,2,-1,2)$ and $\vec{w}_2 = (-3,2,4,1,-2)$.

$$W^{\perp} = \{ \vec{x} \in \mathbb{R}^5 \mid \vec{x} \cdot \begin{bmatrix} 1 \\ 0 \\ 2 \\ -1 \\ 2 \end{bmatrix} = 0 \quad \text{and} \quad \vec{x} \cdot \begin{bmatrix} -3 \\ 2 \\ 4 \\ 1 \\ -2 \end{bmatrix} = 0 \}$$

If we let $\vec{x} = (x_1, x_2, x_3, x_4, x_5)$, we get two equations from these dot products.

$$x_1 + 2x_3 - x_4 + 2x_5 = 0$$

$$-3x_1 + 2x_2 + 4x_3 + x_4 - 2x_5 = 0$$

Put these equations into an augmented matrix,

$$\left[\begin{array}{ccccc|c} 1 & 0 & 2 & -1 & 2 & 0 \\ -3 & 2 & 4 & 1 & -2 & 0 \end{array}\right]$$

then put it into reduced row-echelon form.

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$$\begin{bmatrix} 1 & 0 & 2 & -1 & 2 & | & 0 \\ 0 & 2 & 10 & -2 & 4 & | & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 2 & -1 & 2 & | & 0 \\ 0 & 1 & 5 & -1 & 2 & | & 0 \end{bmatrix}$$

The rref form gives the system of equations

$$x_1 + 2x_3 - x_4 + 2x_5 = 0$$

$$x_2 + 5x_3 - x_4 + 2x_5 = 0$$

which we can solve for the pivot variables.

$$x_1 = -2x_3 + x_4 - 2x_5$$

$$x_2 = -5x_3 + x_4 - 2x_5$$

So we could also express the system as

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = x_3 \begin{bmatrix} -2 \\ -5 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -2 \\ -2 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Which means the orthogonal complement is

$$W^{\perp} = \text{Span}\left(\begin{bmatrix} -2 \\ -5 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ -2 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right)$$

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ORTHOGONAL COMPLEMENTS OF THE FUNDAMENTAL SUBSPACES

1. For the matrix M, find the dimensions of all four fundamental subspaces.

$$M = \begin{bmatrix} -2 & 6 & 0 \\ -1 & 4 & 3 \\ 2 & -5 & 3 \end{bmatrix}$$

Solution:

Put M into reduced row-echelon form.

$$\begin{bmatrix} -2 & 6 & 0 \\ -1 & 4 & 3 \\ 2 & -5 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -3 & 0 \\ -1 & 4 & 3 \\ 2 & -5 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -3 & 0 \\ 0 & 1 & 3 \\ 2 & -5 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -3 & 0 \\ 0 & 1 & 3 \\ 0 & 1 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -3 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 9 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

In reduced row-echelon form, we can see that there are two pivots, which means the rank of M is $r = 2$.

The matrix M is a 3×3 matrix, which means there are $m = 3$ rows and $n = 3$ columns. Therefore, the dimensions of the four fundamental subspaces of M are:

Column space, $C(M)$ $\qquad\qquad$ $r = 2$

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Null space, $N(M)$	$n - r = 3 - 2 = 1$
Row space, $C(M^T)$	$r = 2$
Left null space, $N(M^T)$	$m - r = 3 - 2 = 1$

2. For the matrix M, find the dimensions of all four fundamental subspaces.

$$M = \begin{bmatrix} -1 & 0 & 2 & -4 \\ -2 & 3 & -5 & 1 \\ 1 & -2 & 4 & 0 \end{bmatrix}$$

Solution:

Put M into reduced row-echelon form.

$$\begin{bmatrix} -1 & 0 & 2 & -4 \\ -2 & 3 & -5 & 1 \\ 1 & -2 & 4 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -2 & 4 \\ -2 & 3 & -5 & 1 \\ 1 & -2 & 4 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -2 & 4 \\ 0 & 3 & -9 & 9 \\ 1 & -2 & 4 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & -2 & 4 \\ 0 & 3 & -9 & 9 \\ 0 & -2 & 6 & -4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -2 & 4 \\ 0 & 1 & -3 & 3 \\ 0 & -2 & 6 & -4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -2 & 4 \\ 0 & 1 & -3 & 3 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & -2 & 4 \\ 0 & 1 & -3 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -2 & 0 \\ 0 & 1 & -3 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -2 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

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In reduced row-echelon form, we can see that there are three pivots, which means the rank of M is $r = 3$.

The matrix M is a 3×4 matrix, which means there are $m = 3$ rows and $n = 4$ columns. Therefore, the dimensions of the four fundamental subspaces of M are:

Column space, $C(M)$	$r = 3$
Null space, $N(M)$	$n - r = 4 - 3 = 1$
Row space, $C(M^T)$	$r = 3$
Left null space, $N(M^T)$	$m - r = 3 - 3 = 0$

■ 3. For the matrix X, find the dimensions of all four fundamental subspaces.

$$X = \begin{bmatrix} 1 & -2 & 4 \\ -3 & 5 & 0 \\ -1 & 2 & 3 \end{bmatrix}$$

Solution:

Put X into reduced row-echelon form.

$$\begin{bmatrix} 1 & -2 & 4 \\ -3 & 5 & 0 \\ -1 & 2 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 & 4 \\ 0 & -1 & 12 \\ -1 & 2 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 & 4 \\ 0 & -1 & 12 \\ 0 & 0 & 7 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 & 4 \\ 0 & 1 & -12 \\ 0 & 0 & 7 \end{bmatrix}$$

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$$\begin{bmatrix} 1 & 0 & -20 \\ 0 & 1 & -12 \\ 0 & 0 & 7 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -20 \\ 0 & 1 & -12 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -12 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

In reduced row-echelon form, we can see that there are three pivots, which means the rank of X is $r = 3$.

The matrix X is a 3×3 matrix, which means there are $m = 3$ rows and $n = 3$ columns. Therefore, the dimensions of the four fundamental subspaces of X are:

Column space, $C(X)$	$r = 3$
Null space, $N(X)$	$n - r = 3 - 3 = 0$
Row space, $C(X^T)$	$r = 3$
Left null space, $N(X^T)$	$m - r = 3 - 3 = 0$

■ 4. For the matrix A, find the dimensions of all four fundamental subspaces.

$$A = \begin{bmatrix} -1 & -3 & 2 & 1 \\ -2 & -5 & 5 & -1 \\ -3 & -7 & 8 & -3 \end{bmatrix}$$

Solution:

Put A into reduced row-echelon form.

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$$\begin{bmatrix} -1 & -3 & 2 & 1 \\ -2 & -5 & 5 & -1 \\ -3 & -7 & 8 & -3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & -2 & -1 \\ -2 & -5 & 5 & -1 \\ -3 & -7 & 8 & -3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & -2 & -1 \\ 0 & 1 & 1 & -3 \\ -3 & -7 & 8 & -3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 & -2 & -1 \\ 0 & 1 & 1 & -3 \\ 0 & 2 & 2 & -6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -5 & 8 \\ 0 & 1 & 1 & -3 \\ 0 & 2 & 2 & -6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -5 & 8 \\ 0 & 1 & 1 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

In reduced row-echelon form, we can see that there are two pivots, which means the rank of A is $r = 2$.

The matrix A is a 3×4 matrix, which means there are $m = 3$ rows and $n = 4$ columns. Therefore, the dimensions of the four fundamental subspaces of A are:

Column space, $C(A)$	$r = 2$
Null space, $N(A)$	$n - r = 4 - 2 = 2$
Row space, $C(A^T)$	$r = 2$
Left null space, $N(A^T)$	$m - r = 3 - 2 = 1$

■ 5. For the matrix A, find the dimensions of all four fundamental subspaces.

$$A = \begin{bmatrix} 1 & -1 & 3 & 0 & 2 \\ -1 & 4 & -3 & 1 & 0 \\ 2 & -11 & 6 & -3 & -2 \end{bmatrix}$$

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Solution:

Put A into reduced row-echelon form.

$$\begin{bmatrix} 1 & -1 & 3 & 0 & 2 \\ -1 & 4 & -3 & 1 & 0 \\ 2 & -11 & 6 & -3 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 3 & 0 & 2 \\ 0 & 3 & 0 & 1 & 2 \\ 2 & -11 & 6 & -3 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 3 & 0 & 2 \\ 0 & 3 & 0 & 1 & 2 \\ 0 & -9 & 0 & -3 & -6 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 & 3 & 0 & 2 \\ 0 & 1 & 0 & \frac{1}{3} & \frac{2}{3} \\ 0 & -9 & 0 & -3 & -6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 3 & 0 & 2 \\ 0 & 1 & 0 & \frac{1}{3} & \frac{2}{3} \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 3 & \frac{1}{3} & \frac{8}{3} \\ 0 & 1 & 0 & \frac{1}{3} & \frac{2}{3} \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

In reduced row-echelon form, we can see that there are two pivots, which means the rank of A is $r = 2$.

The matrix A is a 3×5 matrix, which means there are $m = 3$ rows and $n = 5$ columns. Therefore, the dimensions of the four fundamental subspaces of A are:

Column space, $C(A)$	$r = 2$
Null space, $N(A)$	$n - r = 5 - 2 = 3$
Row space, $C(A^T)$	$r = 2$
Left null space, $N(A^T)$	$m - r = 3 - 2 = 1$

■ 6. For the matrix M, find the dimensions of all four fundamental subspaces.

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$$M = \begin{bmatrix} -2 & 2 & -4 \\ 1 & -2 & 0 \\ -3 & 5 & -2 \\ 1 & 2 & 8 \end{bmatrix}$$

Solution:

Put M into reduced row-echelon form.

$$\begin{bmatrix} -2 & 2 & -4 \\ 1 & -2 & 0 \\ -3 & 5 & -2 \\ 1 & 2 & 8 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 2 \\ 1 & -2 & 0 \\ -3 & 5 & -2 \\ 1 & 2 & 8 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 2 \\ 0 & -1 & -2 \\ -3 & 5 & -2 \\ 1 & 2 & 8 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 2 \\ 0 & -1 & -2 \\ 0 & 2 & 4 \\ 1 & 2 & 8 \end{bmatrix} \rightarrow$$

$$\begin{bmatrix} 1 & -1 & 2 \\ 0 & -1 & -2 \\ 0 & 2 & 4 \\ 0 & 3 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & 2 \\ 0 & 2 & 4 \\ 0 & 3 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 2 \\ 0 & 2 & 4 \\ 0 & 3 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 3 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

In reduced row-echelon form, we can see that there are two pivots, which means the rank of M is $r = 2$.

The matrix M is a 4×3 matrix, which means there are $m = 4$ rows and $n = 3$ columns. Therefore, the dimensions of the four fundamental subspaces of M are:

Column space, $C(M)$	$r = 2$
Null space, $N(M)$	$n - r = 3 - 2 = 1$
Row space, $C(M^T)$	$r = 2$

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Left null space, $N(M^T)$ $m - r = 4 - 2 = 2$

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PROJECTION ONTO THE SUBSPACE

■ 1. If $\vec{x}$ is a vector in $\mathbb{R}^3$, find an expression for the projection of any $\vec{x}$ onto the subspace V.

$$V = \text{Span}\left(\begin{bmatrix} -1 \\ 3 \\ -2 \end{bmatrix}, \begin{bmatrix} 0 \\ -2 \\ 3 \end{bmatrix} \right)$$

Solution:

Because the vectors that span V are linearly independent, the matrix A of the basis vectors that define V is

$$A = \begin{bmatrix} -1 & 0 \\ 3 & -2 \\ -2 & 3 \end{bmatrix}$$

The transpose A^T is

$$A^T = \begin{bmatrix} -1 & 3 & -2 \\ 0 & -2 & 3 \end{bmatrix}$$

Find $A^T A$.

$$A^T A = \begin{bmatrix} -1 & 3 & -2 \\ 0 & -2 & 3 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 3 & -2 \\ -2 & 3 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} -1(-1) + 3(3) - 2(-2) & -1(0) + 3(-2) - 2(3) \\ 0(-1) - 2(3) + 3(-2) & 0(0) - 2(-2) + 3(3) \end{bmatrix}$$

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$$A^TA = \begin{bmatrix} 1+9+4 & 0-6-6 \\ 0-6-6 & 0+4+9 \end{bmatrix}$$

$$A^TA = \begin{bmatrix} 14 & -12 \\ -12 & 13 \end{bmatrix}$$

Find the inverse of A^TA.

$$[A^TA \mid I_2] = \left[\begin{array}{cc|cc} 14 & -12 & 1 & 0 \\ -12 & 13 & 0 & 1 \end{array}\right]$$

$$[A^TA \mid I_2] = \left[\begin{array}{cc|cc} 1 & -\frac{6}{7} & \frac{1}{14} & 0 \\ -12 & 13 & 0 & 1 \end{array}\right]$$

$$[A^TA \mid I_2] = \left[\begin{array}{cc|cc} 1 & -\frac{6}{7} & \frac{1}{14} & 0 \\ 0 & \frac{19}{7} & \frac{6}{7} & 1 \end{array}\right]$$

$$[A^TA \mid I_2] = \left[\begin{array}{cc|cc} 1 & -\frac{6}{7} & \frac{1}{14} & 0 \\ 0 & 1 & \frac{6}{19} & \frac{7}{19} \end{array}\right]$$

$$[A^TA \mid I_2] = \left[\begin{array}{cc|cc} 1 & 0 & \frac{13}{38} & \frac{6}{19} \\ 0 & 1 & \frac{6}{19} & \frac{7}{19} \end{array}\right]$$

So $(A^TA)^{-1}$ is

$$(A^TA)^{-1} = \begin{bmatrix} \frac{13}{38} & \frac{6}{19} \\ \frac{6}{19} & \frac{7}{19} \end{bmatrix}$$

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$$(A^TA)^{-1} = \frac{1}{38}\begin{bmatrix}13 & 12\\ 12 & 14\end{bmatrix}$$

Now the projection of $\vec{x}$ onto the subspace V will be

$$\text{Proj}_V\vec{x} = A(A^TA)^{-1}A^T\vec{x}$$

$$\text{Proj}_V\vec{x} = \begin{bmatrix}-1 & 0\\ 3 & -2\\ -2 & 3\end{bmatrix}\frac{1}{38}\begin{bmatrix}13 & 12\\ 12 & 14\end{bmatrix}\begin{bmatrix}-1 & 3 & -2\\ 0 & -2 & 3\end{bmatrix}\vec{x}$$

$$\text{Proj}_V\vec{x} = \frac{1}{38}\begin{bmatrix}-1 & 0\\ 3 & -2\\ -2 & 3\end{bmatrix}\begin{bmatrix}13 & 12\\ 12 & 14\end{bmatrix}\begin{bmatrix}-1 & 3 & -2\\ 0 & -2 & 3\end{bmatrix}\vec{x}$$

First, simplify $(A^TA)^{-1}A^T$.

$$\text{Proj}_V\vec{x} = \frac{1}{38}\begin{bmatrix}-1 & 0\\ 3 & -2\\ -2 & 3\end{bmatrix}\begin{bmatrix}13(-1)+12(0) & 13(3)+12(-2) & 13(-2)+12(3)\\ 12(-1)+14(0) & 12(3)+14(-2) & 12(-2)+14(3)\end{bmatrix}\vec{x}$$

$$\text{Proj}_V\vec{x} = \frac{1}{38}\begin{bmatrix}-1 & 0\\ 3 & -2\\ -2 & 3\end{bmatrix}\begin{bmatrix}-13+0 & 39-24 & -26+36\\ -12+0 & 36-28 & -24+42\end{bmatrix}\vec{x}$$

$$\text{Proj}_V\vec{x} = \frac{1}{38}\begin{bmatrix}-1 & 0\\ 3 & -2\\ -2 & 3\end{bmatrix}\begin{bmatrix}-13 & 15 & 10\\ -12 & 8 & 18\end{bmatrix}\vec{x}$$

Next, simplify $A(A^TA)^{-1}A^T$.

$$\text{Proj}_V\vec{x} = \frac{1}{38}\begin{bmatrix}-1(-13)+0(-12) & -1(15)+0(8) & -1(10)+0(18)\\ 3(-13)-2(-12) & 3(15)-2(8) & 3(10)-2(18)\\ -2(-13)+3(-12) & -2(15)+3(8) & -2(10)+3(18)\end{bmatrix}\vec{x}$$

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$$\text{Proj}_V \vec{x} = \frac{1}{38}\begin{bmatrix} 13+0 & -15+0 & -10+0 \\ -39+24 & 45-16 & 30-36 \\ 26-36 & -30+24 & -20+54 \end{bmatrix}\vec{x}$$

$$\text{Proj}_V \vec{x} = \frac{1}{38}\begin{bmatrix} 13 & -15 & -10 \\ -15 & 29 & -6 \\ -10 & -6 & 34 \end{bmatrix}\vec{x}$$

■ 2. If $\vec{x}$ is a vector in $\mathbb{R}^3$, find an expression for the projection of any $\vec{x}$ onto the subspace V.

$$V = \text{Span}\left(\begin{bmatrix} -2 \\ -4 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ -3 \\ -2 \end{bmatrix}\right)$$

Solution:

Because the vectors that span V are linearly independent, the matrix A of the basis vectors that define V is

$$A = \begin{bmatrix} -2 & -1 & 1 \\ -4 & 1 & -3 \\ 0 & 2 & -2 \end{bmatrix}$$

The transpose A^T is

$$A^T = \begin{bmatrix} -2 & -4 & 0 \\ -1 & 1 & 2 \\ 1 & -3 & -2 \end{bmatrix}$$

Find $A^T A$.

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$$A^TA = \begin{bmatrix} -2 & -4 & 0 \\ -1 & 1 & 2 \\ 1 & -3 & -2 \end{bmatrix} \begin{bmatrix} -2 & -1 & 1 \\ -4 & 1 & -3 \\ 0 & 2 & -2 \end{bmatrix}$$

$$A^TA = \begin{bmatrix} -2(-2) - 4(-4) + 0(0) & -2(-1) - 4(1) + 0(2) & -2(1) - 4(-3) + 0(-2) \\ -1(-2) + 1(-4) + 2(0) & -1(-1) + 1(1) + 2(2) & -1(1) + 1(-3) + 2(-2) \\ 1(-2) - 3(-4) - 2(0) & 1(-1) - 3(1) - 2(2) & 1(1) - 3(-3) - 2(-2) \end{bmatrix}$$

$$A^TA = \begin{bmatrix} 4 + 16 + 0 & 2 - 4 + 0 & -2 + 12 + 0 \\ 2 - 4 + 0 & 1 + 1 + 4 & -1 - 3 - 4 \\ -2 + 12 - 0 & -1 - 3 - 4 & 1 + 9 + 4 \end{bmatrix}$$

$$A^TA = \begin{bmatrix} 20 & -2 & 10 \\ -2 & 6 & -8 \\ 10 & -8 & 14 \end{bmatrix}$$

Find the inverse of A^TA.

$$[A^TA \mid I_3] = \left[\begin{array}{ccc|ccc} 20 & -2 & 10 & 1 & 0 & 0 \\ -2 & 6 & -8 & 0 & 1 & 0 \\ 10 & -8 & 14 & 0 & 0 & 1 \end{array}\right]$$

$$[A^TA \mid I_3] = \left[\begin{array}{ccc|ccc} 1 & -\frac{1}{10} & \frac{1}{2} & \frac{1}{20} & 0 & 0 \\ -2 & 6 & -8 & 0 & 1 & 0 \\ 10 & -8 & 14 & 0 & 0 & 1 \end{array}\right]$$

$$[A^TA \mid I_3] = \left[\begin{array}{ccc|ccc} 1 & -\frac{1}{10} & \frac{1}{2} & \frac{1}{20} & 0 & 0 \\ 0 & \frac{29}{5} & -7 & \frac{1}{10} & 1 & 0 \\ 10 & -8 & 14 & 0 & 0 & 1 \end{array}\right]$$

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$$[A^TA \mid I_3] = \left[\begin{array}{ccc|ccc} 1 & -\frac{1}{10} & \frac{1}{2} & \frac{1}{20} & 0 & 0 \\ 0 & \frac{29}{5} & -7 & \frac{1}{10} & 1 & 0 \\ 0 & -7 & 9 & -\frac{1}{2} & 0 & 1 \end{array}\right]$$

$$[A^TA \mid I_3] = \left[\begin{array}{ccc|ccc} 1 & -\frac{1}{10} & \frac{1}{2} & \frac{1}{20} & 0 & 0 \\ 0 & 1 & -\frac{35}{29} & \frac{1}{58} & \frac{5}{29} & 0 \\ 0 & -7 & 9 & -\frac{1}{2} & 0 & 1 \end{array}\right]$$

$$[A^TA \mid I_3] = \left[\begin{array}{ccc|ccc} 1 & 0 & \frac{11}{29} & \frac{3}{58} & \frac{1}{58} & 0 \\ 0 & 1 & -\frac{35}{29} & \frac{1}{58} & \frac{5}{29} & 0 \\ 0 & -7 & 9 & -\frac{1}{2} & 0 & 1 \end{array}\right]$$

$$[A^TA \mid I_3] = \left[\begin{array}{ccc|ccc} 1 & 0 & \frac{11}{29} & \frac{3}{58} & \frac{1}{58} & 0 \\ 0 & 1 & -\frac{35}{29} & \frac{1}{58} & \frac{5}{29} & 0 \\ 0 & 0 & \frac{16}{29} & -\frac{11}{29} & \frac{35}{29} & 1 \end{array}\right]$$

$$[A^TA \mid I_3] = \left[\begin{array}{ccc|ccc} 1 & 0 & \frac{11}{29} & \frac{3}{58} & \frac{1}{58} & 0 \\ 0 & 1 & -\frac{35}{29} & \frac{1}{58} & \frac{5}{29} & 0 \\ 0 & 0 & 1 & -\frac{11}{16} & \frac{35}{16} & \frac{29}{16} \end{array}\right]$$

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$$[A^TA \mid I_3] = \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{5}{16} & -\frac{13}{16} & -\frac{11}{16} \\ 0 & 1 & -\frac{35}{29} & \frac{1}{58} & \frac{5}{29} & 0 \\ 0 & 0 & 1 & -\frac{11}{16} & \frac{35}{16} & \frac{29}{16} \end{array}\right]$$

$$[A^TA \mid I_3] = \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{5}{16} & -\frac{13}{16} & -\frac{11}{16} \\ 0 & 1 & 0 & -\frac{13}{16} & \frac{45}{16} & \frac{35}{16} \\ 0 & 0 & 1 & -\frac{11}{16} & \frac{35}{16} & \frac{29}{16} \end{array}\right]$$

So $(A^TA)^{-1}$ is

$$(A^TA)^{-1} = \begin{bmatrix} \frac{5}{16} & -\frac{13}{16} & -\frac{11}{16} \\ -\frac{13}{16} & \frac{45}{16} & \frac{35}{16} \\ -\frac{11}{16} & \frac{35}{16} & \frac{29}{16} \end{bmatrix}$$

$$(A^TA)^{-1} = \frac{1}{16}\begin{bmatrix} 5 & -13 & -11 \\ -13 & 45 & 35 \\ -11 & 35 & 29 \end{bmatrix}$$

Then the projection of $\vec{x}$ onto the subspace V will be

$$\text{Proj}_V \vec{x} = A(A^TA)^{-1}A^T\vec{x}$$

$$\text{Proj}_V \vec{x} = \begin{bmatrix} -2 & -1 & 1 \\ -4 & 1 & -3 \\ 0 & 2 & -2 \end{bmatrix} \frac{1}{16} \begin{bmatrix} 5 & -13 & -11 \\ -13 & 45 & 35 \\ -11 & 35 & 29 \end{bmatrix} \begin{bmatrix} -2 & -4 & 0 \\ -1 & 1 & 2 \\ 1 & -3 & -2 \end{bmatrix} \vec{x}$$

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$$\text{Proj}_V \vec{x} = \frac{1}{16}\begin{bmatrix}-2 & -1 & 1\\ -4 & 1 & -3\\ 0 & 2 & -2\end{bmatrix}\begin{bmatrix}5 & -13 & -11\\ -13 & 45 & 35\\ -11 & 35 & 29\end{bmatrix}\begin{bmatrix}-2 & -4 & 0\\ -1 & 1 & 2\\ 1 & -3 & -2\end{bmatrix}\vec{x}$$

First, simplify $(A^TA)^{-1}A^T$.

$$\begin{bmatrix}5(-2)-13(-1)-11(1) & 5(-4)-13(1)-11(-3) & 5(0)-13(2)-11(-2)\\ -13(-2)+45(-1)+35(1) & -13(-4)+45(1)+35(-3) & -13(0)+45(2)+35(-2)\\ -11(-2)+35(-1)+29(1) & -11(-4)+35(1)+29(-3) & -11(0)+35(2)+29(-2)\end{bmatrix}$$

$$\begin{bmatrix}-10+13-11 & -20-13+33 & 0-26+22\\ 26-45+35 & 52+45-105 & 0+90-70\\ 22-35+29 & 44+35-87 & 0+70-58\end{bmatrix}$$

$$\begin{bmatrix}-8 & 0 & -4\\ 16 & -8 & 20\\ 16 & -8 & 12\end{bmatrix}$$

So we get

$$\text{Proj}_V \vec{x} = \frac{1}{16}\begin{bmatrix}-2 & -1 & 1\\ -4 & 1 & -3\\ 0 & 2 & -2\end{bmatrix}\begin{bmatrix}-8 & 0 & -4\\ 16 & -8 & 20\\ 16 & -8 & 12\end{bmatrix}\vec{x}$$

Now, simplify $A(A^TA)^{-1}A^T$.

$$\text{Proj}_V \vec{x} = \frac{1}{4}\begin{bmatrix}-2 & -1 & 1\\ -4 & 1 & -3\\ 0 & 2 & -2\end{bmatrix}\begin{bmatrix}-2 & 0 & -1\\ 4 & -2 & 5\\ 4 & -2 & 3\end{bmatrix}\vec{x}$$

$$\text{Proj}_V \vec{x} = \frac{1}{4}\begin{bmatrix}-2(-2)-1(4)+1(4) & -2(0)-1(-2)+1(-2) & -2(-1)-1(5)+1(3)\\ -4(-2)+1(4)-3(4) & -4(0)+1(-2)-3(-2) & -4(-1)+1(5)-3(3)\\ 0(-2)+2(4)-2(4) & 0(0)+2(-2)-2(-2) & 0(-1)+2(5)-2(3)\end{bmatrix}\vec{x}$$

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$$\text{Proj}_V \vec{x} = \frac{1}{4}\begin{bmatrix} 4-4+4 & 0+2-2 & 2-5+3 \\ 8+4-12 & 0-2+6 & 4+5-9 \\ 0+8-8 & 0-4+4 & 0+10-6 \end{bmatrix} \vec{x}$$

$$\text{Proj}_V \vec{x} = \frac{1}{4}\begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix} \vec{x}$$

$$\text{Proj}_V \vec{x} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \vec{x}$$

■ 3. If $\vec{x}$ is a vector in $\mathbb{R}^3$, find an expression for the projection of any $\vec{x}$ onto the subspace S, if S is spanned by $\vec{x}_1$ and $\vec{x}_2$.

$$\vec{x}_1 \begin{bmatrix} 1 \\ -3 \\ 1 \end{bmatrix} \text{ and } \vec{x}_2 \begin{bmatrix} -1 \\ 0 \\ -2 \end{bmatrix}$$

Solution:

Because the vectors that span S are linearly independent, the matrix A of the basis vectors that define S is

$$A = \begin{bmatrix} 1 & -1 \\ -3 & 0 \\ 1 & -2 \end{bmatrix}$$

The transpose A^T is

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$$A^T = \begin{bmatrix} 1 & -3 & 1 \\ -1 & 0 & -2 \end{bmatrix}$$

Find A^TA.

$$A^TA = \begin{bmatrix} 1 & -3 & 1 \\ -1 & 0 & -2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -3 & 0 \\ 1 & -2 \end{bmatrix}$$

$$A^TA = \begin{bmatrix} 1(1) - 3(-3) + 1(1) & 1(-1) - 3(0) + 1(-2) \\ -1(1) + 0(-3) - 2(1) & -1(-1) + 0(0) - 2(-2) \end{bmatrix}$$

$$A^TA = \begin{bmatrix} 1 + 9 + 1 & -1 - 0 - 2 \\ -1 + 0 - 2 & 1 + 0 + 4 \end{bmatrix}$$

$$A^TA = \begin{bmatrix} 11 & -3 \\ -3 & 5 \end{bmatrix}$$

Find the inverse of A^TA.

$$[A^TA \mid I_2] = \left[\begin{array}{cc|cc} 11 & -3 & 1 & 0 \\ -3 & 5 & 0 & 1 \end{array}\right]$$

$$[A^TA \mid I_2] = \left[\begin{array}{cc|cc} 1 & -\frac{3}{11} & \frac{1}{11} & 0 \\ -3 & 5 & 0 & 1 \end{array}\right]$$

$$[A^TA \mid I_2] = \left[\begin{array}{cc|cc} 1 & -\frac{3}{11} & \frac{1}{11} & 0 \\ 0 & \frac{46}{11} & \frac{3}{11} & 1 \end{array}\right]$$

$$[A^TA \mid I_2] = \left[\begin{array}{cc|cc} 1 & -\frac{3}{11} & \frac{1}{11} & 0 \\ 0 & 1 & \frac{3}{46} & \frac{11}{46} \end{array}\right]$$

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$$[A^TA \mid I_2] = \left[\begin{array}{cc|cc} 1 & 0 & \frac{5}{46} & \frac{3}{46} \\ 0 & 1 & \frac{3}{46} & \frac{11}{46} \end{array}\right]$$

So $(A^TA)^{-1}$ is

$$(A^TA)^{-1} = \begin{bmatrix} \frac{5}{46} & \frac{3}{46} \\ \frac{3}{46} & \frac{11}{46} \end{bmatrix}$$

$$(A^TA)^{-1} = \frac{1}{46}\begin{bmatrix} 5 & 3 \\ 3 & 11 \end{bmatrix}$$

The projection of $\vec{x}$ onto the subspace S will be

$$\text{Proj}_S\vec{x} = A(A^TA)^{-1}A^T\vec{x}$$

$$\text{Proj}_S\vec{x} = \begin{bmatrix} 1 & -1 \\ -3 & 0 \\ 1 & -2 \end{bmatrix}\frac{1}{46}\begin{bmatrix} 5 & 3 \\ 3 & 11 \end{bmatrix}\begin{bmatrix} 1 & -3 & 1 \\ -1 & 0 & -2 \end{bmatrix}\vec{x}$$

$$\text{Proj}_S\vec{x} = \frac{1}{46}\begin{bmatrix} 1 & -1 \\ -3 & 0 \\ 1 & -2 \end{bmatrix}\begin{bmatrix} 5 & 3 \\ 3 & 11 \end{bmatrix}\begin{bmatrix} 1 & -3 & 1 \\ -1 & 0 & -2 \end{bmatrix}\vec{x}$$

First, simplify $(A^TA)^{-1}A^T$.

$$\text{Proj}_S\vec{x} = \frac{1}{46}\begin{bmatrix} 1 & -1 \\ -3 & 0 \\ 1 & -2 \end{bmatrix}\begin{bmatrix} 5(1)+3(-1) & 5(-3)+3(0) & 5(1)+3(-2) \\ 3(1)+11(-1) & 3(-3)+11(0) & 3(1)+11(-2) \end{bmatrix}\vec{x}$$

$$\text{Proj}_S\vec{x} = \frac{1}{46}\begin{bmatrix} 1 & -1 \\ -3 & 0 \\ 1 & -2 \end{bmatrix}\begin{bmatrix} 5-3 & -15+0 & 5-6 \\ 3-11 & -9+0 & 3-22 \end{bmatrix}\vec{x}$$

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$$\text{Proj}_S \vec{x} = \frac{1}{46}\begin{bmatrix} 1 & -1 \\ -3 & 0 \\ 1 & -2 \end{bmatrix}\begin{bmatrix} 2 & -15 & -1 \\ -8 & -9 & -19 \end{bmatrix}\vec{x}$$

Now, simplify $A(A^TA)^{-1}A^T$.

$$\text{Proj}_S \vec{x} = \frac{1}{46}\begin{bmatrix} 1(2)-1(-8) & 1(-15)-1(-9) & 1(-1)-1(-19) \\ -3(2)+0(-8) & -3(-15)+0(-9) & -3(-1)+0(-19) \\ 1(2)-2(-8) & 1(-15)-2(-9) & 1(-1)-2(-19) \end{bmatrix}\vec{x}$$

$$\text{Proj}_S \vec{x} = \frac{1}{46}\begin{bmatrix} 2+8 & -15+9 & -1+19 \\ -6+0 & 45+0 & 3+0 \\ 2+16 & -15+18 & -1+38 \end{bmatrix}\vec{x}$$

$$\text{Proj}_S \vec{x} = \frac{1}{46}\begin{bmatrix} 10 & -6 & 18 \\ -6 & 45 & 3 \\ 18 & 3 & 37 \end{bmatrix}\vec{x}$$

■ 4. If $\vec{x}$ is a vector in $\mathbb{R}^4$, find an expression for the projection of any $\vec{x}$ onto the subspace S, if S is spanned by $\vec{x}_1$ and $\vec{x}_2$.

$$\vec{x}_1 = \frac{1}{2}\begin{bmatrix} 1 \\ -2 \\ -1 \\ -1 \end{bmatrix} \text{ and } \vec{x}_2 = \frac{1}{2}\begin{bmatrix} -1 \\ 0 \\ 1 \\ -3 \end{bmatrix}$$

Solution:

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Because the vectors that span S are linearly independent, the matrix A of the basis vectors that define S is

$$A = \frac{1}{2}\begin{bmatrix} 1 & -1 \\ -2 & 0 \\ -1 & 1 \\ -1 & -3 \end{bmatrix}$$

The transpose A^T is

$$A^T = \frac{1}{2}\begin{bmatrix} 1 & -2 & -1 & -1 \\ -1 & 0 & 1 & -3 \end{bmatrix}$$

Find $A^T A$.

$$A^T A = \frac{1}{2}\begin{bmatrix} 1 & -2 & -1 & -1 \\ -1 & 0 & 1 & -3 \end{bmatrix} \frac{1}{2}\begin{bmatrix} 1 & -1 \\ -2 & 0 \\ -1 & 1 \\ -1 & -3 \end{bmatrix}$$

$$A^T A = \frac{1}{4}\begin{bmatrix} 1 & -2 & -1 & -1 \\ -1 & 0 & 1 & -3 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -2 & 0 \\ -1 & 1 \\ -1 & -3 \end{bmatrix}$$

$$A^T A = \frac{1}{4}\begin{bmatrix} 1(1) - 2(-2) - 1(-1) - 1(-1) & 1(-1) - 2(0) - 1(1) - 1(-3) \\ -1(1) + 0(-2) + 1(-1) - 3(-1) & -1(-1) + 0(0) + 1(1) - 3(-3) \end{bmatrix}$$

$$A^T A = \frac{1}{4}\begin{bmatrix} 1 + 4 + 1 + 1 & -1 - 0 - 1 + 3 \\ -1 + 0 - 1 + 3 & 1 + 0 + 1 + 9 \end{bmatrix}$$

$$A^T A = \frac{1}{4}\begin{bmatrix} 7 & 1 \\ 1 & 11 \end{bmatrix}$$

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$$A^TA = \begin{bmatrix} \frac{7}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{11}{4} \end{bmatrix}$$

Find the inverse of A^TA.

$$[A^TA \mid I_2] = \left[\begin{array}{cc|cc} \frac{7}{4} & \frac{1}{4} & 1 & 0 \\ \frac{1}{4} & \frac{11}{4} & 0 & 1 \end{array}\right]$$

$$[A^TA \mid I_2] = \left[\begin{array}{cc|cc} 1 & \frac{1}{7} & \frac{4}{7} & 0 \\ \frac{1}{4} & \frac{11}{4} & 0 & 1 \end{array}\right]$$

$$[A^TA \mid I_2] = \left[\begin{array}{cc|cc} 1 & \frac{1}{7} & \frac{4}{7} & 0 \\ 0 & \frac{19}{7} & -\frac{1}{7} & 1 \end{array}\right]$$

$$[A^TA \mid I_2] = \left[\begin{array}{cc|cc} 1 & \frac{1}{7} & \frac{4}{7} & 0 \\ 0 & 1 & -\frac{1}{19} & \frac{7}{19} \end{array}\right]$$

$$[A^TA \mid I_2] = \left[\begin{array}{cc|cc} 1 & 0 & \frac{11}{19} & -\frac{1}{19} \\ 0 & 1 & -\frac{1}{19} & \frac{7}{19} \end{array}\right]$$

So $(A^TA)^{-1}$ is

$$(A^TA)^{-1} = \begin{bmatrix} \frac{11}{19} & -\frac{1}{19} \\ -\frac{1}{19} & \frac{7}{19} \end{bmatrix}$$

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$$(A^TA)^{-1} = \frac{1}{19}\begin{bmatrix} 11 & -1 \\ -1 & 7 \end{bmatrix}$$

The projection of $\vec{x}$ onto the subspace S will be

$$\text{Proj}_S\vec{x} = A(A^TA)^{-1}A^T\vec{x}$$

$$\text{Proj}_S\vec{x} = \frac{1}{2}\begin{bmatrix} 1 & -1 \\ -2 & 0 \\ -1 & 1 \\ -1 & -3 \end{bmatrix}\frac{1}{19}\begin{bmatrix} 11 & -1 \\ -1 & 7 \end{bmatrix}\frac{1}{2}\begin{bmatrix} 1 & -2 & -1 & -1 \\ -1 & 0 & 1 & -3 \end{bmatrix}\vec{x}$$

First, simplify $(A^TA)^{-1}A^T$.

$$\text{Proj}_S\vec{x} = \frac{1}{76}\begin{bmatrix} 1 & -1 \\ -2 & 0 \\ -1 & 1 \\ -1 & -3 \end{bmatrix}\begin{bmatrix} 11(1)-1(-1) & 11(-2)-1(0) & 11(-1)-1(1) & 11(-1)-1(-3) \\ -1(1)+7(-1) & -1(-2)+7(0) & -1(-1)+7(1) & -1(-1)+7(-3) \end{bmatrix}\vec{x}$$

$$\text{Proj}_S\vec{x} = \frac{1}{76}\begin{bmatrix} 1 & -1 \\ -2 & 0 \\ -1 & 1 \\ -1 & -3 \end{bmatrix}\begin{bmatrix} 11+1 & -22-0 & -11-1 & -11+3 \\ -1-7 & 2+0 & 1+7 & 1-21 \end{bmatrix}\vec{x}$$

$$\text{Proj}_S\vec{x} = \frac{1}{76}\begin{bmatrix} 1 & -1 \\ -2 & 0 \\ -1 & 1 \\ -1 & -3 \end{bmatrix}\begin{bmatrix} 12 & -22 & -12 & -8 \\ -8 & 2 & 8 & -20 \end{bmatrix}\vec{x}$$

Now, simplify $A(A^TA)^{-1}A^T$.

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$$\text{Proj}_S \vec{x} = \frac{1}{76}\begin{bmatrix} 1(12)-1(-8) & 1(-22)-1(2) & 1(-12)-1(8) & 1(-8)-1(-20) \\ -2(12)+0(-8) & -2(-22)+0(2) & -2(-12)+0(8) & -2(-8)+0(-20) \\ -1(12)+1(-8) & -1(-22)+1(2) & -1(-12)+1(8) & -1(-8)+1(-20) \\ -1(12)-3(-8) & -1(-22)-3(2) & -1(-12)-3(8) & -1(-8)-3(-20) \end{bmatrix}\vec{x}$$

$$\text{Proj}_S \vec{x} = \frac{1}{76}\begin{bmatrix} 12+8 & -22-2 & -12-8 & -8+20 \\ -24+0 & 44+0 & 24+0 & 16+0 \\ -12-8 & 22+2 & 12+8 & 8-20 \\ -12+24 & 22-6 & 12-24 & 8+60 \end{bmatrix}\vec{x}$$

$$\text{Proj}_S \vec{x} = \frac{1}{76}\begin{bmatrix} 20 & -24 & -20 & 12 \\ -24 & 44 & 24 & 16 \\ -20 & 24 & 20 & -12 \\ 12 & 16 & -12 & 68 \end{bmatrix}\vec{x}$$

$$\text{Proj}_S \vec{x} = \frac{1}{19}\begin{bmatrix} 5 & -6 & -5 & 3 \\ -6 & 11 & 6 & 4 \\ -5 & 6 & 5 & -3 \\ 3 & 4 & -3 & 17 \end{bmatrix}\vec{x}$$

■ 5. If $\vec{x}$ is a vector in $\mathbb{R}^4$, find an expression for the projection of any $\vec{x}$ onto the subspace V.

$$V = \text{Span}\left(\begin{bmatrix} -1 \\ 0 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 1 \\ 1 \end{bmatrix}\right)$$

Solution:

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Because the vectors that span V are linearly independent, the matrix A of the basis vectors that define V is

$$A = \begin{bmatrix} -1 & 1 & 0 \\ 0 & 3 & 2 \\ 2 & -1 & 1 \\ -1 & 2 & 1 \end{bmatrix}$$

The transpose A^T is

$$A^T = \begin{bmatrix} -1 & 0 & 2 & -1 \\ 1 & 3 & -1 & 2 \\ 0 & 2 & 1 & 1 \end{bmatrix}$$

Find $A^T A$.

$$A^T A = \begin{bmatrix} -1 & 0 & 2 & -1 \\ 1 & 3 & -1 & 2 \\ 0 & 2 & 1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 1 & 0 \\ 0 & 3 & 2 \\ 2 & -1 & 1 \\ -1 & 2 & 1 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} -1(-1) + 0(0) + 2(2) - 1(-1) & -1(1) + 0(3) + 2(-1) - 1(2) & -1(0) + 0(2) + 2(1) - 1(1) \\ 1(-1) + 3(0) - 1(2) + 2(-1) & 1(1) + 3(3) - 1(-1) + 2(2) & 1(0) + 3(2) - 1(1) + 2(1) \\ 0(-1) + 2(0) + 1(2) + 1(-1) & 0(1) + 2(3) + 1(-1) + 1(2) & 0(0) + 2(2) + 1(1) + 1(1) \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 1 + 0 + 4 + 1 & -1 + 0 - 2 - 2 & 0 + 0 + 2 - 1 \\ -1 + 0 - 2 - 2 & 1 + 9 + 1 + 4 & 0 + 6 - 1 + 2 \\ 0 + 0 + 2 - 1 & 0 + 6 - 1 + 2 & 0 + 4 + 1 + 1 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 6 & -5 & 1 \\ -5 & 15 & 7 \\ 1 & 7 & 6 \end{bmatrix}$$

Find the inverse of $A^T A$.

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$$[A^TA \mid I_3] = \left[\begin{array}{ccc|ccc} 6 & -5 & 1 & 1 & 0 & 0 \\ -5 & 15 & 7 & 0 & 1 & 0 \\ 1 & 7 & 6 & 0 & 0 & 1 \end{array}\right]$$

$$[A^TA \mid I_3] = \left[\begin{array}{ccc|ccc} 1 & -\frac{5}{6} & \frac{1}{6} & \frac{1}{6} & 0 & 0 \\ -5 & 15 & 7 & 0 & 1 & 0 \\ 1 & 7 & 6 & 0 & 0 & 1 \end{array}\right]$$

$$[A^TA \mid I_3] = \left[\begin{array}{ccc|ccc} 1 & -\frac{5}{6} & \frac{1}{6} & \frac{1}{6} & 0 & 0 \\ 0 & \frac{65}{6} & \frac{47}{6} & \frac{5}{6} & 1 & 0 \\ 1 & 7 & 6 & 0 & 0 & 1 \end{array}\right]$$

$$[A^TA \mid I_3] = \left[\begin{array}{ccc|ccc} 1 & -\frac{5}{6} & \frac{1}{6} & \frac{1}{6} & 0 & 0 \\ 0 & \frac{65}{6} & \frac{47}{6} & \frac{5}{6} & 1 & 0 \\ 0 & \frac{47}{6} & \frac{35}{6} & -\frac{1}{6} & 0 & 1 \end{array}\right]$$

$$[A^TA \mid I_3] = \left[\begin{array}{ccc|ccc} 1 & -\frac{5}{6} & \frac{1}{6} & \frac{1}{6} & 0 & 0 \\ 0 & 1 & \frac{47}{65} & \frac{1}{13} & \frac{6}{65} & 0 \\ 0 & \frac{47}{6} & \frac{35}{6} & -\frac{1}{6} & 0 & 1 \end{array}\right]$$

$$[A^TA \mid I_3] = \left[\begin{array}{ccc|ccc} 1 & 0 & \frac{10}{13} & \frac{3}{13} & \frac{1}{13} & 0 \\ 0 & 1 & \frac{47}{65} & \frac{1}{13} & \frac{6}{65} & 0 \\ 0 & \frac{47}{6} & \frac{35}{6} & -\frac{1}{6} & 0 & 1 \end{array}\right]$$

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$$[A^TA \mid I_3] = \begin{bmatrix} 1 & 0 & \frac{10}{13} & \mid & \frac{3}{13} & \frac{1}{13} & 0 \\ 0 & 1 & \frac{47}{65} & \mid & \frac{1}{13} & \frac{6}{65} & 0 \\ 0 & 0 & \frac{11}{65} & \mid & -\frac{10}{13} & -\frac{47}{65} & 1 \end{bmatrix}$$

$$[A^TA \mid I_3] = \begin{bmatrix} 1 & 0 & \frac{10}{13} & \mid & \frac{3}{13} & \frac{1}{13} & 0 \\ 0 & 1 & \frac{47}{65} & \mid & \frac{1}{13} & \frac{6}{65} & 0 \\ 0 & 0 & 1 & \mid & -\frac{50}{11} & -\frac{47}{11} & \frac{65}{11} \end{bmatrix}$$

$$[A^TA \mid I_3] = \begin{bmatrix} 1 & 0 & 0 & \mid & \frac{41}{11} & \frac{37}{11} & -\frac{50}{11} \\ 0 & 1 & \frac{47}{65} & \mid & \frac{1}{13} & \frac{6}{65} & 0 \\ 0 & 0 & 1 & \mid & -\frac{50}{11} & -\frac{47}{11} & \frac{65}{11} \end{bmatrix}$$

$$[A^TA \mid I_3] = \begin{bmatrix} 1 & 0 & 0 & \mid & \frac{41}{11} & \frac{37}{11} & -\frac{50}{11} \\ 0 & 1 & 0 & \mid & \frac{37}{11} & \frac{35}{11} & -\frac{47}{11} \\ 0 & 0 & 1 & \mid & -\frac{50}{11} & -\frac{47}{11} & \frac{65}{11} \end{bmatrix}$$

So $(A^TA)^{-1}$ is

$$(A^TA)^{-1} = \begin{bmatrix} \frac{41}{11} & \frac{37}{11} & -\frac{50}{11} \\ \frac{37}{11} & \frac{35}{11} & -\frac{47}{11} \\ -\frac{50}{11} & -\frac{47}{11} & \frac{65}{11} \end{bmatrix}$$

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$$(A^TA)^{-1} = \frac{1}{11}\begin{bmatrix} 41 & 37 & -50 \\ 37 & 35 & -47 \\ -50 & -47 & 65 \end{bmatrix}$$

Then the projection of $\vec{x}$ onto the subspace V will be

$$\text{Proj}_V \vec{x} = A(A^TA)^{-1}A^T\vec{x}$$

$$\text{Proj}_V \vec{x} = \begin{bmatrix} -1 & 1 & 0 \\ 0 & 3 & 2 \\ 2 & -1 & 1 \\ -1 & 2 & 1 \end{bmatrix} \frac{1}{11}\begin{bmatrix} 41 & 37 & -50 \\ 37 & 35 & -47 \\ -50 & -47 & 65 \end{bmatrix} \begin{bmatrix} -1 & 0 & 2 & -1 \\ 1 & 3 & -1 & 2 \\ 0 & 2 & 1 & 1 \end{bmatrix} \vec{x}$$

$$\text{Proj}_V \vec{x} = \frac{1}{11}\begin{bmatrix} -1 & 1 & 0 \\ 0 & 3 & 2 \\ 2 & -1 & 1 \\ -1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 41 & 37 & -50 \\ 37 & 35 & -47 \\ -50 & -47 & 65 \end{bmatrix} \begin{bmatrix} -1 & 0 & 2 & -1 \\ 1 & 3 & -1 & 2 \\ 0 & 2 & 1 & 1 \end{bmatrix} \vec{x}$$

First, simplify $(A^TA)^{-1}A^T$.

$$\begin{bmatrix} 41(-1)+37(1)-50(0) & 41(0)+37(3)-50(2) & 41(2)+37(-1)-50(1) & 41(-1)+37(2)-50(1) \\ 37(-1)+35(1)-47(0) & 37(0)+35(3)-47(2) & 37(2)+35(-1)-47(1) & 37(-1)+35(2)-47(1) \\ -50(-1)-47(1)+65(0) & -50(0)-47(3)+65(2) & -50(2)-47(-1)+65(1) & -50(-1)-47(2)+65(1) \end{bmatrix}$$

$$\begin{bmatrix} -41+37-0 & 0+111-100 & 82-37-50 & -41+74-50 \\ -37+35-0 & 0+105-94 & 74-35-47 & -37+70-47 \\ 50-47+0 & 0-141+130 & -100+47+65 & 50-94+65 \end{bmatrix}$$

$$\begin{bmatrix} -4 & 11 & -5 & -17 \\ -2 & 11 & -8 & -14 \\ 3 & -11 & 12 & 21 \end{bmatrix}$$

So we get

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$$\text{Proj}_V \vec{x} = \frac{1}{11}\begin{bmatrix} -1 & 1 & 0 \\ 0 & 3 & 2 \\ 2 & -1 & 1 \\ -1 & 2 & 1 \end{bmatrix}\begin{bmatrix} -4 & 11 & -5 & -17 \\ -2 & 11 & -8 & -14 \\ 3 & -11 & 12 & 21 \end{bmatrix}\vec{x}$$

Next, simplify $A(A^TA)^{-1}A^T$.

$$\frac{1}{11}\begin{bmatrix} -1(-4)+1(-2)+0(3) & -1(11)+1(11)+0(-11) & -1(-5)+1(-8)+0(12) & -1(-17)+1(-14)+0(21) \\ 0(-4)+3(-2)+2(3) & 0(11)+3(11)+2(-11) & 0(-5)+3(-8)+2(12) & 0(-17)+3(-14)+2(21) \\ 2(-4)-1(-2)+1(3) & 2(11)-1(11)+1(-11) & 2(-5)-1(-8)+1(12) & 2(-17)-1(-14)+1(21) \\ -1(-4)+2(-2)+1(3) & -1(11)+2(11)+1(-11) & -1(-5)+2(-8)+1(12) & -1(-17)+2(-14)+1(21) \end{bmatrix}$$

$$\frac{1}{11}\begin{bmatrix} 4-2+0 & -11+11+0 & 5-8+0 & 17-14+0 \\ 0-6+6 & 0+33-22 & 0-24+24 & 0-42+42 \\ -8+2+3 & 22-11-11 & -10+8+12 & -34+14+21 \\ 4-4+3 & -11+22-11 & 5-16+12 & 17-28+21 \end{bmatrix}$$

$$\frac{1}{11}\begin{bmatrix} 2 & 0 & -3 & 3 \\ 0 & 11 & 0 & 0 \\ -3 & 0 & 10 & 1 \\ 3 & 0 & 1 & 10 \end{bmatrix}$$

So we get

$$\text{Proj}_V \vec{x} = \frac{1}{11}\begin{bmatrix} 2 & 0 & -3 & 3 \\ 0 & 11 & 0 & 0 \\ -3 & 0 & 10 & 1 \\ 3 & 0 & 1 & 10 \end{bmatrix}\vec{x}$$

■ 6. If $\vec{x}$ is a vector in $\mathbb{R}^4$, find an expression for the projection of any $\vec{x}$ onto the subspace S, if S is spanned by $\vec{x}_1$ and $\vec{x}_2$.

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$$\vec{x}_1 = \frac{1}{2}\begin{bmatrix} 2 \\ 8 \\ -4 \end{bmatrix} \text{ and } \vec{x}_2 = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$$

Solution:

Because the vectors that span S are linearly independent, the matrix A of the basis vectors that define S is

$$A = \begin{bmatrix} 1 & -2 \\ 4 & 1 \\ -2 & 0 \end{bmatrix}$$

The transpose A^T is

$$A^T = \begin{bmatrix} 1 & 4 & -2 \\ -2 & 1 & 0 \end{bmatrix}$$

Find $A^T A$.

$$A^T A = \begin{bmatrix} 1 & 4 & -2 \\ -2 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 4 & 1 \\ -2 & 0 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 1(1) + 4(4) - 2(-2) & 1(-2) + 4(1) - 2(0) \\ -2(1) + 1(4) + 0(-2) & -2(-2) + 1(1) + 0(0) \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 1 + 16 + 4 & -2 + 4 - 0 \\ -2 + 4 + 0 & 4 + 1 + 0 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 21 & 2 \\ 2 & 5 \end{bmatrix}$$

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Find the inverse of A^TA.

$$[A^TA \mid I_2] = \left[\begin{array}{cc|cc} 21 & 2 & 1 & 0 \\ 2 & 5 & 0 & 1 \end{array}\right]$$

$$[A^TA \mid I_2] = \left[\begin{array}{cc|cc} 1 & \frac{2}{21} & \frac{1}{21} & 0 \\ 2 & 5 & 0 & 1 \end{array}\right]$$

$$[A^TA \mid I_2] = \left[\begin{array}{cc|cc} 1 & \frac{2}{21} & \frac{1}{21} & 0 \\ 0 & \frac{101}{21} & -\frac{2}{21} & 1 \end{array}\right]$$

$$[A^TA \mid I_2] = \left[\begin{array}{cc|cc} 1 & \frac{2}{21} & \frac{1}{21} & 0 \\ 0 & 1 & -\frac{2}{101} & \frac{21}{101} \end{array}\right]$$

$$[A^TA \mid I_2] = \left[\begin{array}{cc|cc} 1 & 0 & \frac{5}{101} & -\frac{2}{101} \\ 0 & 1 & -\frac{2}{101} & \frac{21}{101} \end{array}\right]$$

So $(A^TA)^{-1}$ is

$$(A^TA)^{-1} = \begin{bmatrix} \frac{5}{101} & -\frac{2}{101} \\ -\frac{2}{101} & \frac{21}{101} \end{bmatrix}$$

$$(A^TA)^{-1} = \frac{1}{101}\begin{bmatrix} 5 & -2 \\ -2 & 21 \end{bmatrix}$$

The projection of $\vec{x}$ onto the subspace S will be

$$\text{Proj}_S \vec{x} = A(A^TA)^{-1}A^T\vec{x}$$

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$$\text{Proj}_S \vec{x} = \begin{bmatrix} 1 & -2 \\ 4 & 1 \\ -2 & 0 \end{bmatrix} \frac{1}{101} \begin{bmatrix} 5 & -2 \\ -2 & 21 \end{bmatrix} \begin{bmatrix} 1 & 4 & -2 \\ -2 & 1 & 0 \end{bmatrix} \vec{x}$$

$$\text{Proj}_S \vec{x} = \frac{1}{101} \begin{bmatrix} 1 & -2 \\ 4 & 1 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} 5 & -2 \\ -2 & 21 \end{bmatrix} \begin{bmatrix} 1 & 4 & -2 \\ -2 & 1 & 0 \end{bmatrix} \vec{x}$$

First, simplify $(A^TA)^{-1}A^T$.

$$\text{Proj}_S \vec{x} = \frac{1}{101} \begin{bmatrix} 1 & -2 \\ 4 & 1 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} 5(1)-2(-2) & 5(4)-2(1) & 5(-2)-2(0) \\ -2(1)+21(-2) & -2(4)+21(1) & -2(-2)+21(0) \end{bmatrix} \vec{x}$$

$$\text{Proj}_S \vec{x} = \frac{1}{101} \begin{bmatrix} 1 & -2 \\ 4 & 1 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} 5+4 & 20-2 & -10-0 \\ -2-42 & -8+21 & 4+0 \end{bmatrix} \vec{x}$$

$$\text{Proj}_S \vec{x} = \frac{1}{101} \begin{bmatrix} 1 & -2 \\ 4 & 1 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} 9 & 18 & -10 \\ -44 & 13 & 4 \end{bmatrix} \vec{x}$$

Now, simplify $A(A^TA)^{-1}A^T$.

$$\text{Proj}_S \vec{x} = \frac{1}{101} \begin{bmatrix} 1(9)-2(-44) & 1(18)-2(13) & 1(-10)-2(4) \\ 4(9)+1(-44) & 4(18)+1(13) & 4(-10)+1(4) \\ -2(9)+0(-44) & -2(18)+0(13) & -2(-10)+0(4) \end{bmatrix} \vec{x}$$

$$\text{Proj}_S \vec{x} = \frac{1}{101} \begin{bmatrix} 9+88 & 18-26 & -10-8 \\ 36-44 & 72+13 & -40+4 \\ -18+0 & -36+0 & 20+0 \end{bmatrix} \vec{x}$$

$$\text{Proj}_S \vec{x} = \frac{1}{101} \begin{bmatrix} 97 & -8 & -18 \\ -8 & 85 & -36 \\ -18 & -36 & 20 \end{bmatrix} \vec{x}$$

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LEAST SQUARES SOLUTION

1. Find the least squares solution to the system.

$$x = 2$$

$$x - y = 2$$

$$x + y = 3$$

Solution:

Put each line into slope-intercept form,

$$x = 2$$

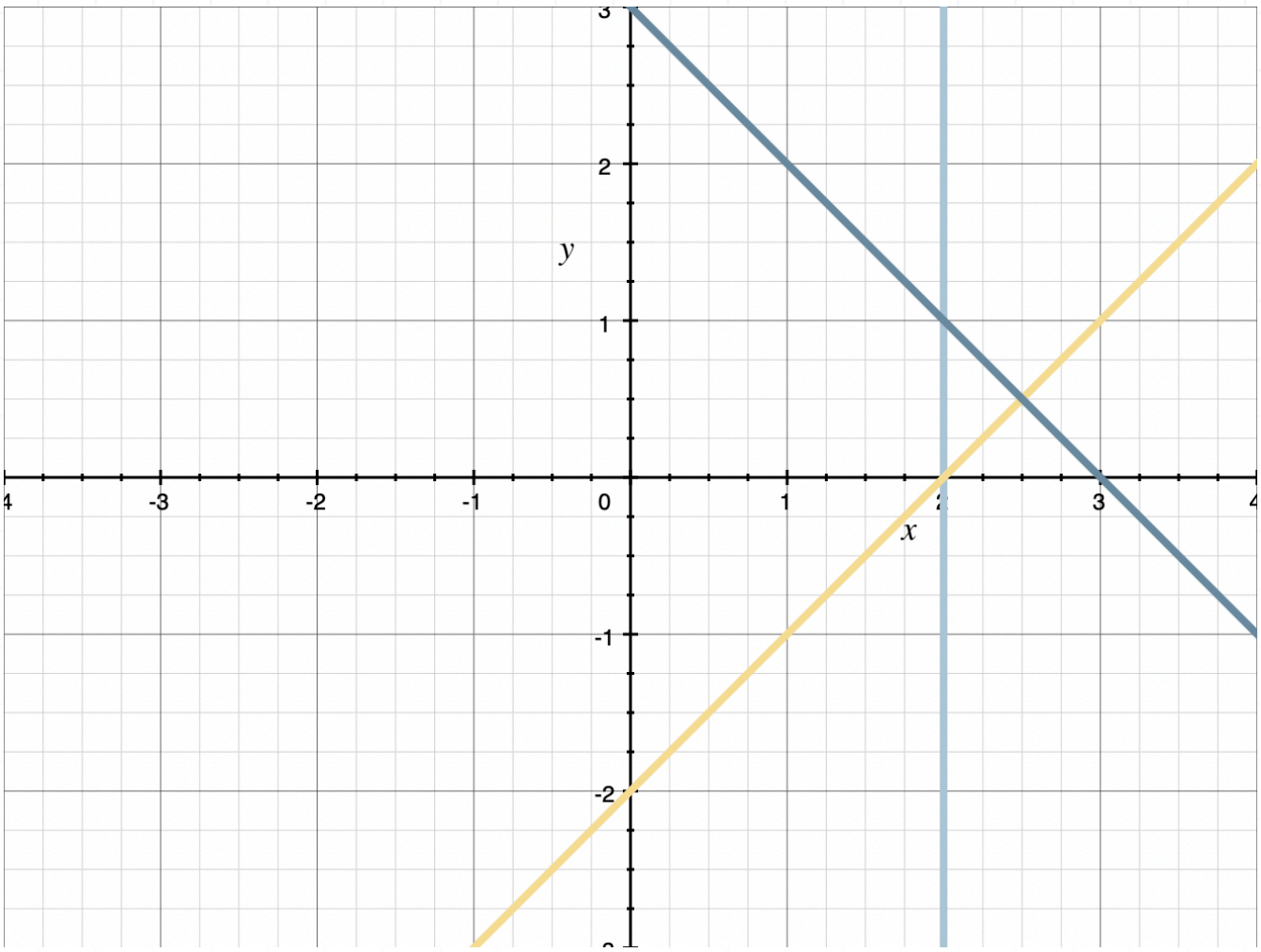
$$y = x - 2$$

$$y = -x + 3$$

then graph all three in the same plane.

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While there are three points where some of the lines intersect, there's no single point where all three lines intersect, which means there's no solution to $A\vec{x} = \vec{b}$.

$$\begin{bmatrix} 1 & 0 \\ 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}$$

In other words, $\vec{b} = (2,2,3)$ is not in the column space of the coefficient matrix A, and there's no vector $\vec{x} = (x, y)$ you can find that makes the $A\vec{x} = \vec{b}$ equation true.

The next best thing we can do is find the least squares solution. By building the matrix equation, we've already found

$$A = \begin{bmatrix} 1 & 0 \\ 1 & -1 \\ 1 & 1 \end{bmatrix}$$

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Now we'll find A^T.

$$A^T = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & 1 \end{bmatrix}$$

Then $A^T A$ is

$$A^T A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & -1 \\ 1 & 1 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 1(1) + 1(1) + 1(1) & 1(0) + 1(-1) + 1(1) \\ 0(1) - 1(1) + 1(1) & 0(0) - 1(-1) + 1(1) \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 1 + 1 + 1 & 0 - 1 + 1 \\ 0 - 1 + 1 & 0 + 1 + 1 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$$

And $A^T \vec{b}$ is

$$A^T \vec{b} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}$$

$$A^T \vec{b} = \begin{bmatrix} 1(2) + 1(2) + 1(3) \\ 0(2) - 1(2) + 1(3) \end{bmatrix}$$

$$A^T \vec{b} = \begin{bmatrix} 2 + 2 + 3 \\ 0 - 2 + 3 \end{bmatrix}$$

$$A^T \vec{b} = \begin{bmatrix} 7 \\ 1 \end{bmatrix}$$

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Then we get

$$A^T A\vec{x}^* = A^T\vec{b}$$

$$\begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}\vec{x}^* = \begin{bmatrix} 7 \\ 1 \end{bmatrix}$$

Then to find $\vec{x}^*$, we'll put the augmented matrix into reduced row-echelon form.

$$\begin{bmatrix} 3 & 0 & | & 7 \\ 0 & 2 & | & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & | & \frac{7}{3} \\ 0 & 2 & | & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & | & \frac{7}{3} \\ 0 & 1 & | & \frac{1}{2} \end{bmatrix}$$

Then the least squares solution is given by the augmented matrix as

$$\vec{x}^* = \left(\frac{7}{3}, \frac{1}{2}\right)$$

2. Find the least squares solution to the system.

$$-x + 2y = 6$$

$$3x + 2y = 0$$

$$y - 3x = -2$$

Solution:

Put each line into slope-intercept form,

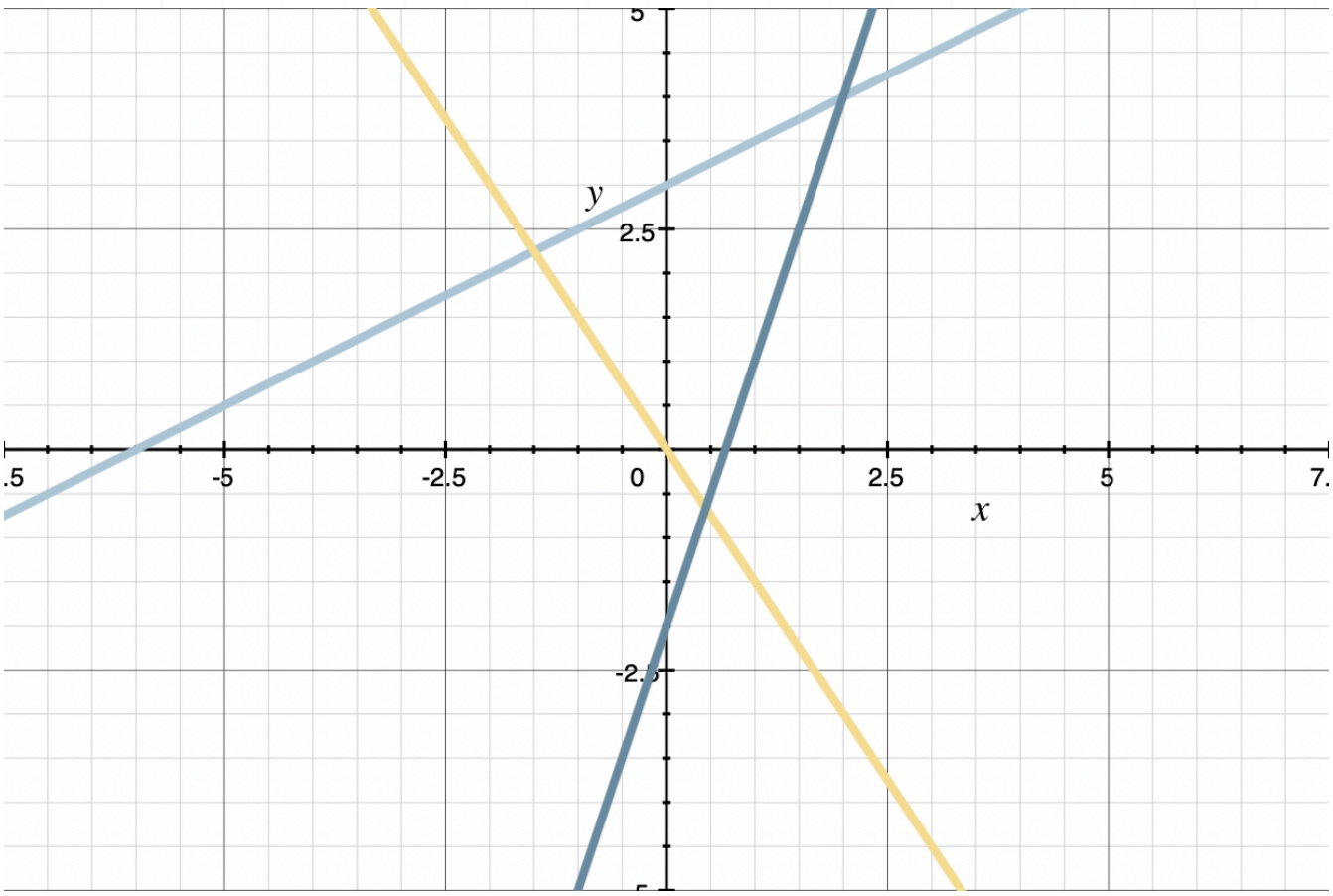
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$$y = \frac{1}{2}x + 3$$

$$y = -\frac{3}{2}x$$

$$y = 3x - 2$$

then graph all three in the same plane.



While there are three points where some of the lines intersect, there's no single point where all three lines intersect, which means there's no solution to $A\vec{x} = \vec{b}$.

$$\begin{bmatrix} -1 & 2 \\ 3 & 2 \\ -3 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 6 \\ 0 \\ -2 \end{bmatrix}$$

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In other words, $\vec{b} = (6,0,-2)$ is not in the column space of the coefficient matrix A, and there's no vector $\vec{x} = (x, y)$ you can find that makes the $A\vec{x} = \vec{b}$ equation true.

The next best thing we can do is find the least squares solution. By building the matrix equation, we've already found

$$A = \begin{bmatrix} -1 & 2 \\ 3 & 2 \\ -3 & 1 \end{bmatrix}$$

Now we'll find A^T.

$$A^T = \begin{bmatrix} -1 & 3 & -3 \\ 2 & 2 & 1 \end{bmatrix}$$

Then $A^T A$ is

$$A^T A = \begin{bmatrix} -1 & 3 & -3 \\ 2 & 2 & 1 \end{bmatrix} \begin{bmatrix} -1 & 2 \\ 3 & 2 \\ -3 & 1 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} -1(-1) + 3(3) - 3(-3) & -1(2) + 3(2) - 3(1) \\ 2(-1) + 2(3) + 1(-3) & 2(2) + 2(2) + 1(1) \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 1 + 9 + 9 & -2 + 6 - 3 \\ -2 + 6 - 3 & 4 + 4 + 1 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 19 & 1 \\ 1 & 9 \end{bmatrix}$$

And $A^T\vec{b}$ is

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$$A^T\vec{b} = \begin{bmatrix} -1 & 3 & -3 \\ 2 & 2 & 1 \end{bmatrix} \begin{bmatrix} 6 \\ 0 \\ -2 \end{bmatrix}$$

$$A^T\vec{b} = \begin{bmatrix} -1(6) + 3(0) - 3(-2) \\ 2(6) + 2(0) + 1(-2) \end{bmatrix}$$

$$A^T\vec{b} = \begin{bmatrix} -6 + 0 + 6 \\ 12 + 0 - 2 \end{bmatrix}$$

$$A^T\vec{b} = \begin{bmatrix} 0 \\ 10 \end{bmatrix}$$

Then we get

$$A^T A\vec{x}^* = A^T\vec{b}$$

$$\begin{bmatrix} 19 & 1 \\ 1 & 9 \end{bmatrix} \vec{x}^* = \begin{bmatrix} 0 \\ 10 \end{bmatrix}$$

Then to find $\vec{x}^*$, we'll put the augmented matrix into reduced row-echelon form.

$$\begin{bmatrix} 19 & 1 & | & 0 \\ 1 & 9 & | & 10 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & \frac{1}{19} & | & 0 \\ 1 & 9 & | & 10 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & \frac{1}{19} & | & 0 \\ 0 & \frac{170}{19} & | & 10 \end{bmatrix}$$

$$\begin{bmatrix} 1 & \frac{1}{19} & | & 0 \\ 0 & 1 & | & \frac{19}{17} \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & | & -\frac{1}{17} \\ 0 & 1 & | & \frac{19}{17} \end{bmatrix}$$

Then the least squares solution is given by the augmented matrix as

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$$\vec{x}^* = \left(-\frac{1}{17}, \frac{19}{17}\right)$$

■ 3. Find the least squares solution to the system.

$$y - 2x = 5$$

$$3x + y = -2$$

$$2x - 4y = 5$$

Solution:

Put each line into slope-intercept form,

$$y = 2x + 5$$

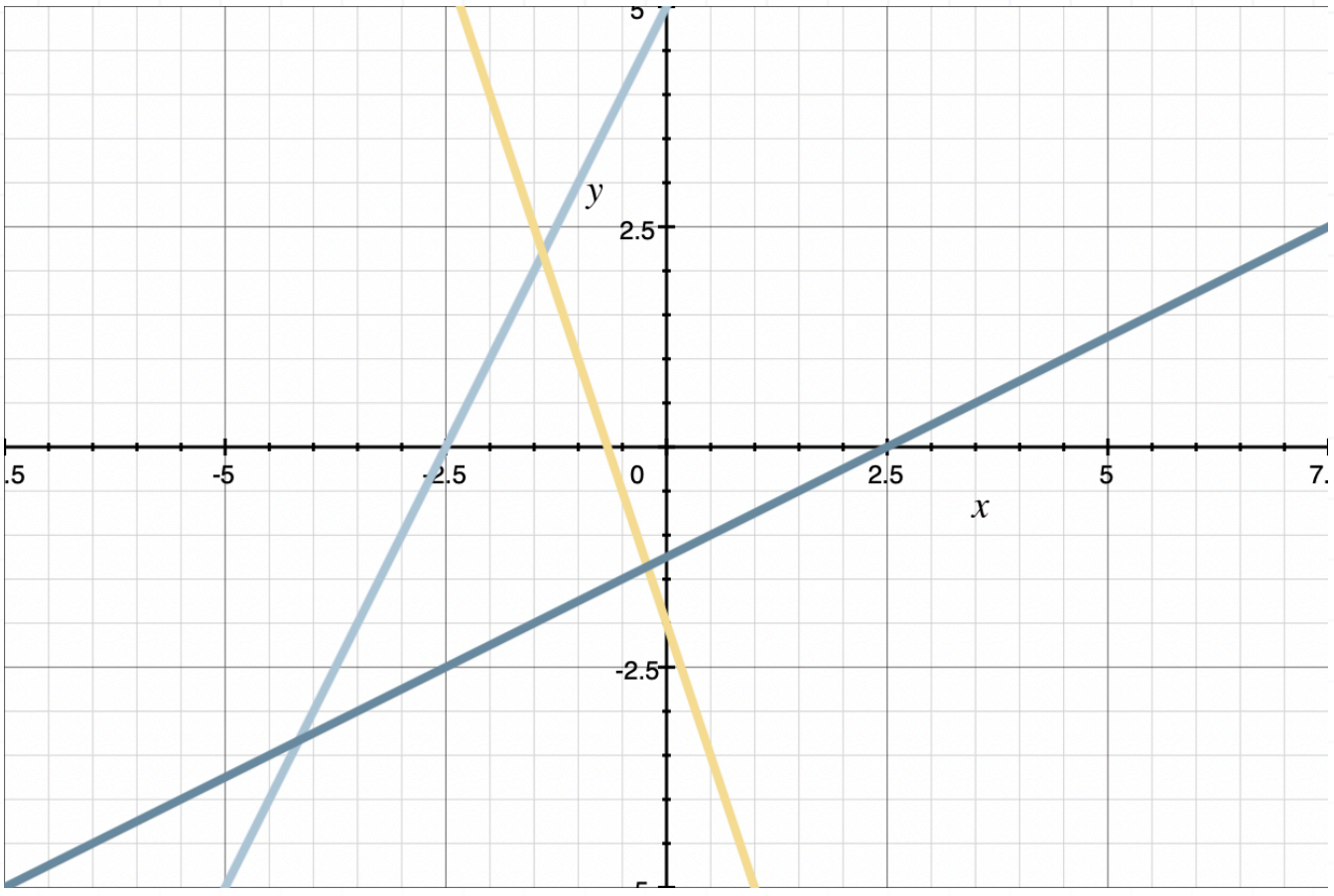
$$y = -3x - 2$$

$$y = \frac{1}{2}x - \frac{5}{4}$$

then graph all three in the same plane.

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While there are three points where some of the lines intersect, there's no single point where all three lines intersect, which means there's no solution to $A\vec{x} = \vec{b}$.

$$\begin{bmatrix} -2 & 1 \\ 3 & 1 \\ 2 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \\ 5 \end{bmatrix}$$

In other words, $\vec{b} = (5, -2, 5)$ is not in the column space of the coefficient matrix A, and there's no vector $\vec{x} = (x, y)$ you can find that makes the $A\vec{x} = \vec{b}$ equation true.

The next best thing we can do is find the least squares solution. By building the matrix equation, we've already found

$$A = \begin{bmatrix} -2 & 1 \\ 3 & 1 \\ 2 & -4 \end{bmatrix}$$

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Now we'll find A^T.

$$A^T = \begin{bmatrix} -2 & 3 & 2 \\ 1 & 1 & -4 \end{bmatrix}$$

Then $A^T A$ is

$$A^T A = \begin{bmatrix} -2 & 3 & 2 \\ 1 & 1 & -4 \end{bmatrix} \begin{bmatrix} -2 & 1 \\ 3 & 1 \\ 2 & -4 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} -2(-2) + 3(3) + 2(2) & -2(1) + 3(1) + 2(-4) \\ 1(-2) + 1(3) - 4(2) & 1(1) + 1(1) - 4(-4) \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 4 + 9 + 4 & -2 + 3 - 8 \\ -2 + 3 - 8 & 1 + 1 + 16 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 17 & -7 \\ -7 & 18 \end{bmatrix}$$

And $A^T \vec{b}$ is

$$A^T \vec{b} = \begin{bmatrix} -2 & 3 & 2 \\ 1 & 1 & -4 \end{bmatrix} \begin{bmatrix} 5 \\ -2 \\ 5 \end{bmatrix}$$

$$A^T \vec{b} = \begin{bmatrix} -2(5) + 3(-2) + 2(5) \\ 1(5) + 1(-2) - 4(5) \end{bmatrix}$$

$$A^T \vec{b} = \begin{bmatrix} -10 - 6 + 10 \\ 5 - 2 - 20 \end{bmatrix}$$

$$A^T \vec{b} = \begin{bmatrix} -6 \\ -17 \end{bmatrix}$$

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Then we get

$$A^T A\vec{x}^* = A^T\vec{b}$$

$$\begin{bmatrix} 17 & -7 \\ -7 & 18 \end{bmatrix}\vec{x}^* = \begin{bmatrix} -6 \\ -17 \end{bmatrix}$$

Then to find $\vec{x}^*$, we'll put the augmented matrix into reduced row-echelon form.

$$\begin{bmatrix} 17 & -7 & | & -6 \\ -7 & 18 & | & -17 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -\frac{7}{17} & | & -\frac{6}{17} \\ -7 & 18 & | & -17 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -\frac{7}{17} & | & -\frac{6}{17} \\ 0 & \frac{257}{17} & | & -\frac{331}{17} \end{bmatrix}$$

$$\begin{bmatrix} 1 & -\frac{7}{17} & | & -\frac{6}{17} \\ 0 & 1 & | & -\frac{331}{257} \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & | & -\frac{227}{257} \\ 0 & 1 & | & -\frac{331}{257} \end{bmatrix}$$

Then the least squares solution is given by the augmented matrix as

$$\vec{x}^* = \left(-\frac{227}{257}, -\frac{331}{257}\right)$$

■ 4. Find the least squares solution to the system.

$$y - 3x = 5$$

$$x + y = -3$$

$$2x - 2y = 3$$

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Solution:

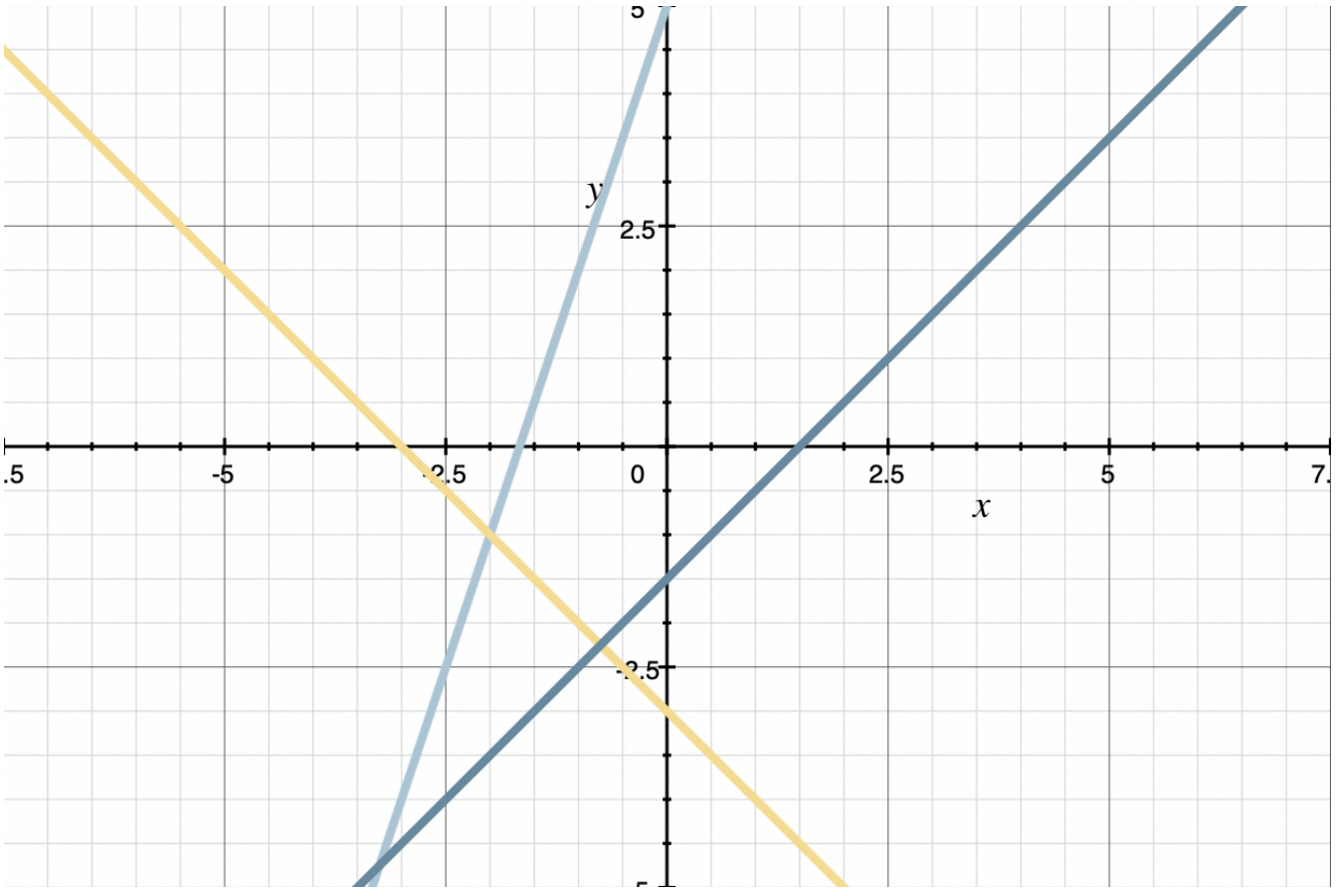
Put each line into slope-intercept form,

$$y = 3x + 5$$

$$y = -x - 3$$

$$y = x - \frac{3}{2}$$

then graph all three in the same plane.



While there are three points where some of the lines intersect, there's no single point where all three lines intersect, which means there's no solution to $A\vec{x} = \vec{b}$.

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$$\begin{bmatrix} -3 & 1 \\ 1 & 1 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ -3 \\ 3 \end{bmatrix}$$

In other words, $\vec{b} = (5, -3, 3)$ is not in the column space of the coefficient matrix A, and there's no vector $\vec{x} = (x, y)$ you can find that makes the $A\vec{x} = \vec{b}$ equation true.

The next best thing we can do is find the least squares solution. By building the matrix equation, we've already found

$$A = \begin{bmatrix} -3 & 1 \\ 1 & 1 \\ 2 & -2 \end{bmatrix}$$

Now we'll find A^T.

$$A^T = \begin{bmatrix} -3 & 1 & 2 \\ 1 & 1 & -2 \end{bmatrix}$$

Then $A^T A$ is

$$A^T A = \begin{bmatrix} -3 & 1 & 2 \\ 1 & 1 & -2 \end{bmatrix} \begin{bmatrix} -3 & 1 \\ 1 & 1 \\ 2 & -2 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} -3(-3) + 1(1) + 2(2) & -3(1) + 1(1) + 2(-2) \\ 1(-3) + 1(1) - 2(2) & 1(1) + 1(1) - 2(-2) \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 9 + 1 + 4 & -3 + 1 - 4 \\ -3 + 1 - 4 & 1 + 1 + 4 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 14 & -6 \\ -6 & 6 \end{bmatrix}$$

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And $A^T\vec{b}$ is

$$A^T\vec{b} = \begin{bmatrix} -3 & 1 & 2 \\ 1 & 1 & -2 \end{bmatrix} \begin{bmatrix} 5 \\ -3 \\ 3 \end{bmatrix}$$

$$A^T\vec{b} = \begin{bmatrix} -3(5) + 1(-3) + 2(3) \\ 1(5) + 1(-3) - 2(3) \end{bmatrix}$$

$$A^T\vec{b} = \begin{bmatrix} -15 - 3 + 6 \\ 5 - 3 - 6 \end{bmatrix}$$

$$A^T\vec{b} = \begin{bmatrix} -12 \\ -4 \end{bmatrix}$$

Then we get

$$A^T A\vec{x}^* = A^T\vec{b}$$

$$\begin{bmatrix} 14 & -6 \\ -6 & 6 \end{bmatrix} \vec{x}^* = \begin{bmatrix} -12 \\ -4 \end{bmatrix}$$

Then to find $\vec{x}^*$, we'll put the augmented matrix into reduced row-echelon form.

$$\begin{bmatrix} 14 & -6 & | & -12 \\ -6 & 6 & | & -4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -\frac{3}{7} & | & -\frac{6}{7} \\ -6 & 6 & | & -4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -\frac{3}{7} & | & -\frac{6}{7} \\ 0 & \frac{24}{7} & | & -\frac{64}{7} \end{bmatrix}$$

$$\begin{bmatrix} 1 & -\frac{3}{7} & | & -\frac{6}{7} \\ 0 & 1 & | & -\frac{8}{3} \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & | & -2 \\ 0 & 1 & | & -\frac{8}{3} \end{bmatrix}$$

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Then the least squares solution is given by the augmented matrix as

$$\vec{x}^* = \left(-2, -\frac{8}{3}\right)$$

■ 5. Find the least squares solution to the system.

$$2y - 3x = -4$$

$$5x + y = -2$$

$$x + 4y = -1$$

Solution:

Put each line into slope-intercept form,

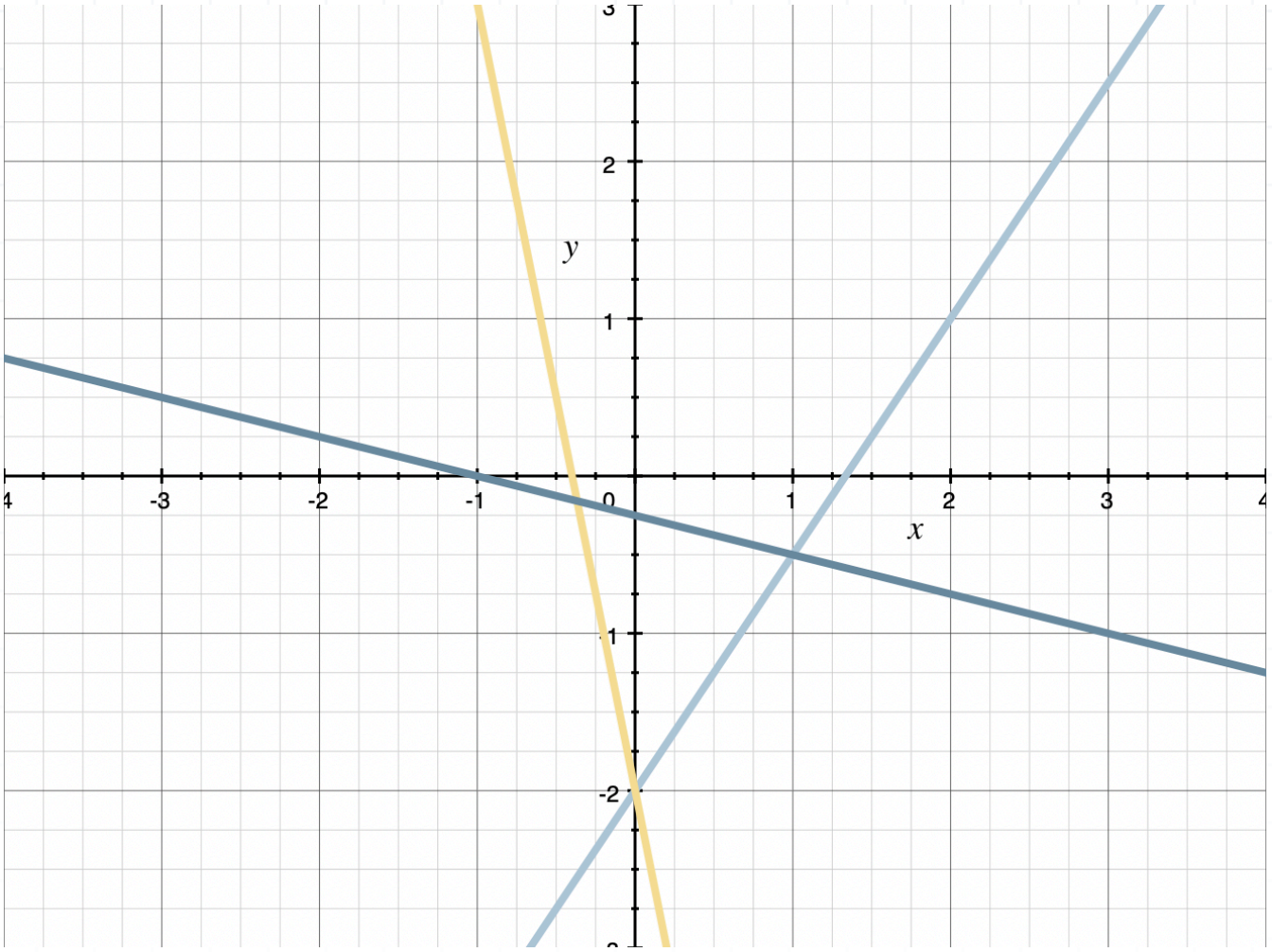
$$y = \frac{3}{2}x - 2$$

$$y = -5x - 2$$

$$y = -\frac{1}{4}x - \frac{1}{4}$$

then graph all three in the same plane.

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While there are three points where some of the lines intersect, there's no single point where all three lines intersect, which means there's no solution to $A\vec{x} = \vec{b}$.

$$\begin{bmatrix} -3 & 2 \\ 5 & 1 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -4 \\ -2 \\ -1 \end{bmatrix}$$

In other words, $\vec{b} = (-4, -2, -1)$ is not in the column space of the coefficient matrix A, and there's no vector $\vec{x} = (x, y)$ you can find that makes the $A\vec{x} = \vec{b}$ equation true.

The next best thing we can do is find the least squares solution. By building the matrix equation, we've already found

$$A = \begin{bmatrix} -3 & 2 \\ 5 & 1 \\ 1 & 4 \end{bmatrix}$$

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Now we'll find A^T.

$$A^T = \begin{bmatrix} -3 & 5 & 1 \\ 2 & 1 & 4 \end{bmatrix}$$

Then $A^T A$ is

$$A^T A = \begin{bmatrix} -3 & 5 & 1 \\ 2 & 1 & 4 \end{bmatrix} \begin{bmatrix} -3 & 2 \\ 5 & 1 \\ 1 & 4 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} -3(-3) + 5(5) + 1(1) & -3(2) + 5(1) + 1(4) \\ 2(-3) + 1(5) + 4(1) & 2(2) + 1(1) + 4(4) \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 9 + 25 + 1 & -6 + 5 + 4 \\ -6 + 5 + 4 & 4 + 1 + 16 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 35 & 3 \\ 3 & 21 \end{bmatrix}$$

And $A^T \vec{b}$ is

$$A^T \vec{b} = \begin{bmatrix} -3 & 5 & 1 \\ 2 & 1 & 4 \end{bmatrix} \begin{bmatrix} -4 \\ -2 \\ -1 \end{bmatrix}$$

$$A^T \vec{b} = \begin{bmatrix} -3(-4) + 5(-2) + 1(-1) \\ 2(-4) + 1(-2) + 4(-1) \end{bmatrix}$$

$$A^T \vec{b} = \begin{bmatrix} 12 - 10 - 1 \\ -8 - 2 - 4 \end{bmatrix}$$

$$A^T \vec{b} = \begin{bmatrix} 1 \\ -14 \end{bmatrix}$$

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Then we get

$$A^T A\vec{x}^* = A^T\vec{b}$$

$$\begin{bmatrix} 35 & 3 \\ 3 & 21 \end{bmatrix}\vec{x}^* = \begin{bmatrix} 1 \\ -14 \end{bmatrix}$$

Then to find $\vec{x}^*$, we'll put the augmented matrix into reduced row-echelon form.

$$\begin{bmatrix} 35 & 3 & | & 1 \\ 3 & 21 & | & -14 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & \frac{3}{35} & | & \frac{1}{35} \\ 3 & 21 & | & -14 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & \frac{3}{35} & | & \frac{1}{35} \\ 0 & \frac{726}{35} & | & -\frac{493}{35} \end{bmatrix}$$

$$\begin{bmatrix} 1 & \frac{3}{35} & | & \frac{1}{35} \\ 0 & 1 & | & -\frac{493}{726} \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & | & \frac{21}{242} \\ 0 & 1 & | & -\frac{493}{726} \end{bmatrix}$$

Then the least squares solution is given by the augmented matrix as

$$\vec{x}^* = \left(\frac{21}{242}, -\frac{493}{726}\right)$$

■ 6. Find the least squares solution to the system.

$$2x - 5y = 4$$

$$x + 6y = 5$$

$$4x - 3y = -6$$

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Solution:

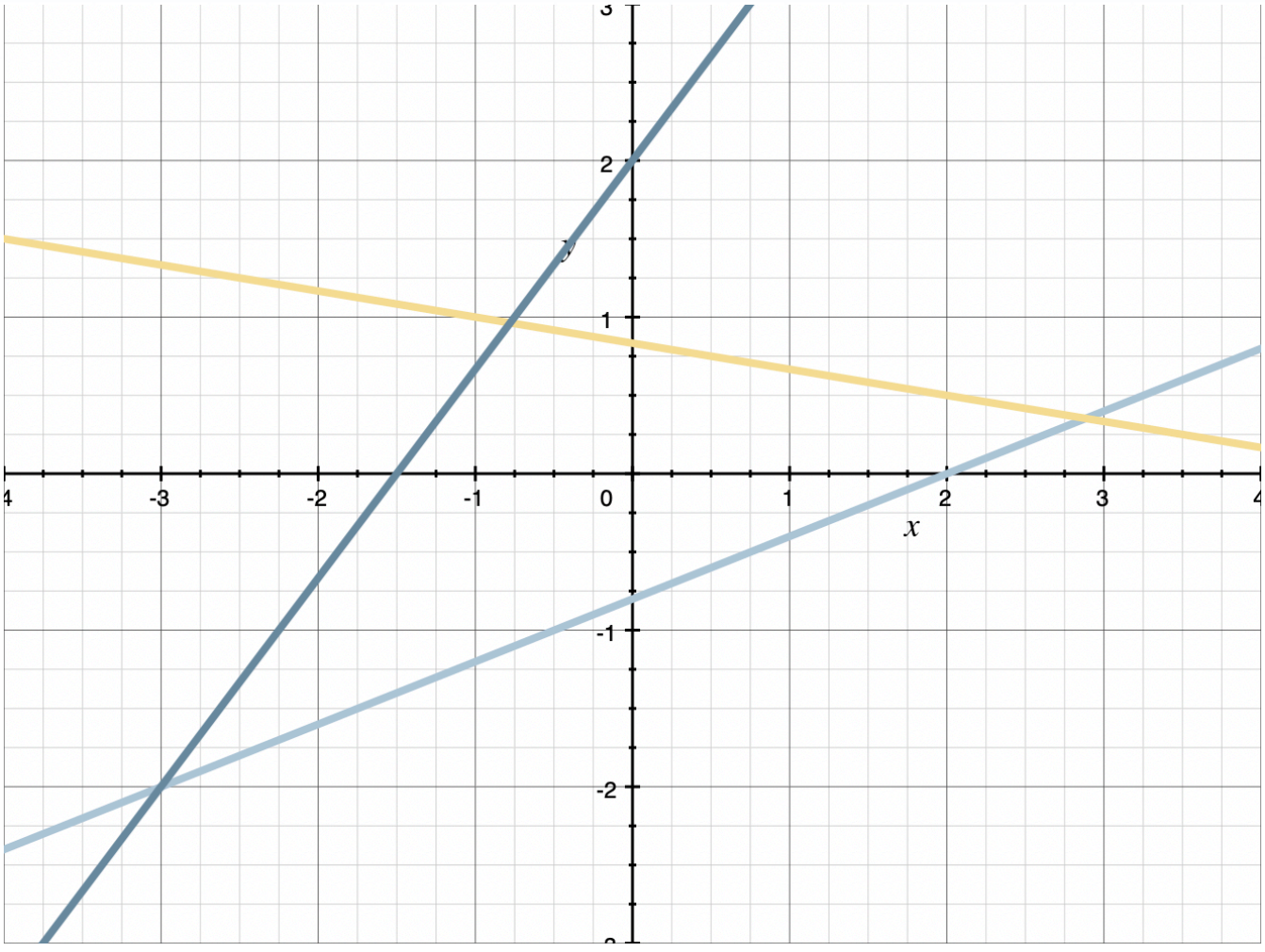
Put each line into slope-intercept form,

$$y = \frac{2}{5}x - \frac{4}{5}$$

$$y = -\frac{1}{6}x + \frac{5}{6}$$

$$y = \frac{4}{3}x + 2$$

then graph all three in the same plane.



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While there are three points where some of the lines intersect, there's no single point where all three lines intersect, which means there's no solution to $A\vec{x} = \vec{b}$.

$$\begin{bmatrix} 2 & -5 \\ 1 & 6 \\ 4 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ -6 \end{bmatrix}$$

In other words, $\vec{b} = (4,5,-6)$ is not in the column space of the coefficient matrix A, and there's no vector $\vec{x} = (x, y)$ you can find that makes the $A\vec{x} = \vec{b}$ equation true.

The next best thing we can do is find the least squares solution. By building the matrix equation, we've already found

$$A = \begin{bmatrix} 2 & -5 \\ 1 & 6 \\ 4 & -3 \end{bmatrix}$$

Now we'll find A^T.

$$A^T = \begin{bmatrix} 2 & 1 & 4 \\ -5 & 6 & -3 \end{bmatrix}$$

Then $A^T A$ is

$$A^T A = \begin{bmatrix} 2 & 1 & 4 \\ -5 & 6 & -3 \end{bmatrix} \begin{bmatrix} 2 & -5 \\ 1 & 6 \\ 4 & -3 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 2(2) + 1(1) + 4(4) & 2(-5) + 1(6) + 4(-3) \\ -5(2) + 6(1) - 3(4) & -5(-5) + 6(6) - 3(-3) \end{bmatrix}$$

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$$A^TA = \begin{bmatrix} 4+1+16 & -10+6-12 \\ -10+6-12 & 25+36+9 \end{bmatrix}$$

$$A^TA = \begin{bmatrix} 21 & -16 \\ -16 & 70 \end{bmatrix}$$

And $A^T\vec{b}$ is

$$A^T\vec{b} = \begin{bmatrix} 2 & 1 & 4 \\ -5 & 6 & -3 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \\ -6 \end{bmatrix}$$

$$A^T\vec{b} = \begin{bmatrix} 2(4)+1(5)+4(-6) \\ -5(4)+6(5)-3(-6) \end{bmatrix}$$

$$A^T\vec{b} = \begin{bmatrix} 8+5-24 \\ -20+30+18 \end{bmatrix}$$

$$A^T\vec{b} = \begin{bmatrix} -11 \\ 28 \end{bmatrix}$$

Then we get

$$A^TA\vec{x}* = A^T\vec{b}$$

$$\begin{bmatrix} 21 & -16 \\ -16 & 70 \end{bmatrix} \vec{x}* = \begin{bmatrix} -11 \\ 28 \end{bmatrix}$$

Then to find $\vec{x}*$, we'll put the augmented matrix into reduced row-echelon form.

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$$\begin{bmatrix} 21 & -16 & | & -11 \\ -16 & 70 & | & 28 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -\frac{16}{21} & | & -\frac{11}{21} \\ -16 & 70 & | & 28 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -\frac{16}{21} & | & -\frac{11}{21} \\ 0 & \frac{1{,}214}{21} & | & \frac{412}{21} \end{bmatrix}$$

$$\begin{bmatrix} 1 & -\frac{16}{21} & | & -\frac{11}{21} \\ 0 & 1 & | & \frac{206}{607} \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & | & -\frac{161}{607} \\ 0 & 1 & | & \frac{206}{607} \end{bmatrix}$$

Then the least squares solution is given by the augmented matrix as

$$\vec{x}^* = \left(-\frac{161}{607}, \frac{206}{607} \right)$$

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COORDINATES IN A NEW BASIS

1. The vectors $\vec{v} = (-2,1)$ and $\vec{w} = (4, -3)$ form an alternate basis for $\mathbb{R}^2$. Use them to transform $\vec{x} = 6\mathbf{i} - 2\mathbf{j}$ into the alternate basis.

Solution:

The vector $\vec{x} = (6, -2)$ is given in terms of the standard basis, and we need to transform it into an alternate basis that's defined by $\vec{v} = (-2,1)$ and $\vec{w} = (4, -3)$.

So let's plug the values we've been given into the matrix equation.

$$A[\vec{x}]_B = \vec{x}$$

$$\begin{bmatrix} -2 & 4 \\ 1 & -3 \end{bmatrix} [\vec{x}]_B = \begin{bmatrix} 6 \\ -2 \end{bmatrix}$$

To find the representation of $\vec{x}$ in the alternate basis, $[\vec{x}]_B$, we'll put the augmented matrix into reduced row-echelon form.

$$\begin{bmatrix} -2 & 4 & | & 6 \\ 1 & -3 & | & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 & | & -3 \\ 1 & -3 & | & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 & | & -3 \\ 0 & -1 & | & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 & | & -3 \\ 0 & 1 & | & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & | & -5 \\ 0 & 1 & | & -1 \end{bmatrix}$$

So $\vec{x} = (6, -2)$, expressed in the alternate basis, is

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$$[\vec{x}]_B = \begin{bmatrix} -5 \\ -1 \end{bmatrix}$$

■ 2. The vectors $\vec{v} = (1, -5)$ and $\vec{w} = (2,4)$ form an alternate basis for $\mathbb{R}^2$. Use them, and an inverse matrix, to transform $\vec{x} = -\mathbf{i}$ into the alternate basis.

Solution:

The vector $\vec{x} = (-1,0)$ is given in terms of the standard basis, and we need to transform it into an alternate basis that's defined by $\vec{v} = (1, -5)$ and $\vec{w} = (2,4)$.

So let's plug the values we've been given into the matrix equation.

$$A[\vec{x}]_B = \vec{x}$$

$$\begin{bmatrix} 1 & 2 \\ -5 & 4 \end{bmatrix}[\vec{x}]_B = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

Find A^{-1} from A.

$$[A \mid I] = \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ -5 & 4 & 0 & 1 \end{array}\right]$$

$$[A \mid I] = \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 14 & 5 & 1 \end{array}\right]$$

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$$[A \mid I] = \begin{bmatrix} 1 & 2 & | & 1 & 0 \\ 0 & 1 & | & \frac{5}{14} & \frac{1}{14} \end{bmatrix}$$

$$[A \mid I] = \begin{bmatrix} 1 & 0 & | & \frac{2}{7} & -\frac{1}{7} \\ 0 & 1 & | & \frac{5}{14} & \frac{1}{14} \end{bmatrix}$$

So the inverse matrix is

$$A^{-1} = \begin{bmatrix} \frac{2}{7} & -\frac{1}{7} \\ \frac{5}{14} & \frac{1}{14} \end{bmatrix}$$

Now to find the representation of $\vec{x} = (-1,0)$ in the alternate basis, we simply multiply the inverse matrix by the vector.

$$[\vec{x}]_B = A^{-1}\vec{x}$$

$$[\vec{x}]_B = \begin{bmatrix} \frac{2}{7} & -\frac{1}{7} \\ \frac{5}{14} & \frac{1}{14} \end{bmatrix} \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

$$[\vec{x}]_B = \begin{bmatrix} \frac{2}{7}(-1) - \frac{1}{7}(0) \\ \frac{5}{14}(-1) + \frac{1}{14}(0) \end{bmatrix}$$

$$[\vec{x}]_B = \begin{bmatrix} -\frac{2}{7} \\ -\frac{5}{14} \end{bmatrix}$$

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■ 3. The vectors $\vec{v} = (-1,0,4)$, $\vec{s} = (2,-3,1)$, and $\vec{w} = (1,-1,2)$ form an alternate basis for $\mathbb{R}^3$. Use them to transform $\vec{x} = -\mathbf{j} + \mathbf{k}$ into the alternate basis.

Solution:

The vector $\vec{x} = (0,-1,1)$ is given in terms of the standard basis, and we need to transform it into an alternate basis that's defined by $\vec{v} = (-1,0,4)$, $\vec{s} = (2,-3,1)$, and $\vec{w} = (1,-1,2)$.

So let's plug the values we've been given into the matrix equation.

$$A[\vec{x}]_B = \vec{x}$$

$$\begin{bmatrix} -1 & 2 & 1 \\ 0 & -3 & -1 \\ 4 & 1 & 2 \end{bmatrix} [\vec{x}]_B = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$$

To find the representation of $\vec{x}$ in the alternate basis, $[\vec{x}]_B$, we'll put the augmented matrix into reduced row-echelon form.

$$\begin{bmatrix} -1 & 2 & 1 & | & 0 \\ 0 & -3 & -1 & | & -1 \\ 4 & 1 & 2 & | & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 & -1 & | & 0 \\ 0 & -3 & -1 & | & -1 \\ 4 & 1 & 2 & | & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 & -1 & | & 0 \\ 0 & -3 & -1 & | & -1 \\ 0 & 9 & 6 & | & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 & -1 & | & 0 \\ 0 & 1 & \frac{1}{3} & | & \frac{1}{3} \\ 0 & 9 & 6 & | & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -\frac{1}{3} & | & \frac{2}{3} \\ 0 & 1 & \frac{1}{3} & | & \frac{1}{3} \\ 0 & 9 & 6 & | & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -\frac{1}{3} & | & \frac{2}{3} \\ 0 & 1 & \frac{1}{3} & | & \frac{1}{3} \\ 0 & 0 & 3 & | & -2 \end{bmatrix}$$

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$$\begin{bmatrix} 1 & 0 & -\frac{1}{3} & | & \frac{2}{3} \\ 0 & 1 & \frac{1}{3} & | & \frac{1}{3} \\ 0 & 0 & 1 & | & -\frac{2}{3} \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -\frac{1}{3} & | & \frac{2}{3} \\ 0 & 1 & 0 & | & \frac{5}{9} \\ 0 & 0 & 1 & | & -\frac{2}{3} \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & | & \frac{4}{9} \\ 0 & 1 & 0 & | & \frac{5}{9} \\ 0 & 0 & 1 & | & -\frac{2}{3} \end{bmatrix}$$

So $\vec{x} = (0, -1, 1)$, expressed in the alternate basis, is

$$[\vec{x}]_B = \begin{bmatrix} \frac{4}{9} \\ \frac{5}{9} \\ -\frac{2}{3} \end{bmatrix}$$

■ 4. The vectors $\vec{v} = (1, -3, 1)$, $\vec{s} = (-3, -3, 2)$, and $\vec{w} = (5, -3, 1)$ form an alternate basis for $\mathbb{R}^3$. Use them, and an inverse matrix to transform $\vec{x} = 2\mathbf{i} + 6\mathbf{j} - \mathbf{k}$ into the alternate basis.

Solution:

The vector $\vec{x} = (2, 6, -1)$ is given in terms of the standard basis, and we need to transform it into an alternate basis that's defined by $\vec{v} = (1, -3, 1)$, $\vec{s} = (-3, -3, 2)$, and $\vec{w} = (5, -3, 1)$.

So let's plug the values we've been given into the matrix equation.

$$A[\vec{x}]_B = \vec{x}$$

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$$\begin{bmatrix} 1 & -3 & 5 \\ -3 & -3 & -3 \\ 1 & 2 & 1 \end{bmatrix} [\vec{x}]_B = \begin{bmatrix} 2 \\ 6 \\ -1 \end{bmatrix}$$

To find the representation of $\vec{x}$ in the alternate basis, $[\vec{x}]_B$, we'll put the augmented matrix into reduced row-echelon form.

Find A^{-1} from A.

$$[A \mid I] = \left[\begin{array}{ccc|ccc} 1 & -3 & 5 & 1 & 0 & 0 \\ -3 & -3 & -3 & 0 & 1 & 0 \\ 1 & 2 & 1 & 0 & 0 & 1 \end{array}\right]$$

$$[A \mid I] = \left[\begin{array}{ccc|ccc} 1 & -3 & 5 & 1 & 0 & 0 \\ 0 & -12 & 12 & 3 & 1 & 0 \\ 1 & 2 & 1 & 0 & 0 & 1 \end{array}\right]$$

$$[A \mid I] = \left[\begin{array}{ccc|ccc} 1 & -3 & 5 & 1 & 0 & 0 \\ 0 & -12 & 12 & 3 & 1 & 0 \\ 0 & 5 & -4 & -1 & 0 & 1 \end{array}\right]$$

$$[A \mid I] = \left[\begin{array}{ccc|ccc} 1 & -3 & 5 & 1 & 0 & 0 \\ 0 & 1 & -1 & -\frac{1}{4} & -\frac{1}{12} & 0 \\ 0 & 5 & -4 & -1 & 0 & 1 \end{array}\right]$$

$$[A \mid I] = \left[\begin{array}{ccc|ccc} 1 & 0 & 2 & \frac{1}{4} & -\frac{1}{4} & 0 \\ 0 & 1 & -1 & -\frac{1}{4} & -\frac{1}{12} & 0 \\ 0 & 5 & -4 & -1 & 0 & 1 \end{array}\right]$$

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$$[A \mid I] = \left[\begin{array}{ccc|ccc} 1 & 0 & 2 & \frac{1}{4} & -\frac{1}{4} & 0 \\ 0 & 1 & -1 & -\frac{1}{4} & -\frac{1}{12} & 0 \\ 0 & 0 & 1 & \frac{1}{4} & \frac{5}{12} & 1 \end{array}\right]$$

$$[A \mid I] = \left[\begin{array}{ccc|ccc} 1 & 0 & 2 & \frac{1}{4} & -\frac{1}{4} & 0 \\ 0 & 1 & 0 & 0 & \frac{1}{3} & 1 \\ 0 & 0 & 1 & \frac{1}{4} & \frac{5}{12} & 1 \end{array}\right]$$

$$[A \mid I] = \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{1}{4} & -\frac{13}{12} & -2 \\ 0 & 1 & 0 & 0 & \frac{1}{3} & 1 \\ 0 & 0 & 1 & \frac{1}{4} & \frac{5}{12} & 1 \end{array}\right]$$

So the inverse matrix is

$$A^{-1} = \begin{bmatrix} -\frac{1}{4} & -\frac{13}{12} & -2 \\ 0 & \frac{1}{3} & 1 \\ \frac{1}{4} & \frac{5}{12} & 1 \end{bmatrix}$$

Now to find the representation of $\vec{x} = (2,6,-1)$ in the alternate basis, we simply multiply the inverse matrix by the vector.

$$[\vec{x}]_B = A^{-1}\vec{x}$$

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$$[\vec{x}]_B = \begin{bmatrix} -\frac{1}{4} & -\frac{13}{12} & -2 \\ 0 & \frac{1}{3} & 1 \\ \frac{1}{4} & \frac{5}{12} & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 6 \\ -1 \end{bmatrix}$$

$$[\vec{x}]_B = \begin{bmatrix} -\frac{1}{4}(2) - \frac{13}{12}(6) - 2(-1) \\ 0(2) + \frac{1}{3}(6) + 1(-1) \\ \frac{1}{4}(2) + \frac{5}{12}(6) + 1(-1) \end{bmatrix}$$

$$[\vec{x}]_B = \begin{bmatrix} -\frac{1}{2} - \frac{13}{2} + 2 \\ 0 + 2 - 1 \\ \frac{1}{2} + \frac{5}{2} - 1 \end{bmatrix}$$

$$[\vec{x}]_B = \begin{bmatrix} -5 \\ 1 \\ 2 \end{bmatrix}$$

■ 5. The vectors $\vec{v} = (-2,3)$ and $\vec{w} = (4,0)$ form an alternate basis for $\mathbb{R}^2$. Use them, and an inverse matrix, to transform $\vec{x} = 6\mathbf{i} - 3\mathbf{j}$ into the alternate basis.

Solution:

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The vector $\vec{x} = (6, -3)$ is given in terms of the standard basis, and we need to transform it into an alternate basis that's defined by $\vec{v} = (-2,3)$ and $\vec{w} = (4,0)$.

So let's plug the values we've been given into the matrix equation.

$$A[\vec{x}]_B = \vec{x}$$

$$\begin{bmatrix} -2 & 4 \\ 3 & 0 \end{bmatrix} [\vec{x}]_B = \begin{bmatrix} 6 \\ -3 \end{bmatrix}$$

Find A^{-1} from A.

$$[A \mid I] = \left[\begin{array}{cc|cc} -2 & 4 & 1 & 0 \\ 3 & 0 & 0 & 1 \end{array}\right]$$

$$[A \mid I] = \left[\begin{array}{cc|cc} 1 & -2 & -\frac{1}{2} & 0 \\ 3 & 0 & 0 & 1 \end{array}\right]$$

$$[A \mid I] = \left[\begin{array}{cc|cc} 1 & -2 & -\frac{1}{2} & 0 \\ 0 & 6 & \frac{3}{2} & 1 \end{array}\right]$$

$$[A \mid I] = \left[\begin{array}{cc|cc} 1 & -2 & -\frac{1}{2} & 0 \\ 0 & 1 & \frac{1}{4} & \frac{1}{6} \end{array}\right]$$

$$[A \mid I] = \left[\begin{array}{cc|cc} 1 & 0 & 0 & \frac{1}{3} \\ 0 & 1 & \frac{1}{4} & \frac{1}{6} \end{array}\right]$$

So the inverse matrix is

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$$A^{-1} = \begin{bmatrix} 0 & \frac{1}{3} \\ \frac{1}{4} & \frac{1}{6} \end{bmatrix}$$

Now to find the representation of $\vec{x} = (6, -3)$ in the alternate basis, we simply multiply the inverse matrix by the vector.

$$[\vec{x}]_B = A^{-1}\vec{x}$$

$$[\vec{x}]_B = \begin{bmatrix} 0 & \frac{1}{3} \\ \frac{1}{4} & \frac{1}{6} \end{bmatrix} \begin{bmatrix} 6 \\ -3 \end{bmatrix}$$

$$[\vec{x}]_B = \begin{bmatrix} 0(6) + \frac{1}{3}(-3) \\ \frac{1}{4}(6) + \frac{1}{6}(-3) \end{bmatrix}$$

$$[\vec{x}]_B = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

■ 6. The vectors $\vec{v} = (-1,3,2)$, $\vec{s} = (-2,4,-4)$, and $\vec{w} = (1,-2,0)$ form an alternate basis for $\mathbb{R}^3$. Use them to transform $\vec{x} = -2\mathbf{i} - 4\mathbf{k}$ into the alternate basis.

Solution:

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The vector $\vec{x} = (-2,0,-4)$ is given in terms of the standard basis, and we need to transform it into an alternate basis that's defined by $\vec{v} = (-1,3,2)$, $\vec{s} = (-2,4,-4)$, and $\vec{w} = (1,-2,0)$.

So let's plug the values we've been given into the matrix equation.

$$A[\vec{x}]_B = \vec{x}$$

$$\begin{bmatrix} -1 & -2 & 1 \\ 3 & 4 & -2 \\ 2 & -4 & 0 \end{bmatrix} [\vec{x}]_B = \begin{bmatrix} -2 \\ 0 \\ -4 \end{bmatrix}$$

To find the representation of $\vec{x}$ in the alternate basis, $[\vec{x}]_B$, we'll put the augmented matrix into reduced row-echelon form.

$$\begin{bmatrix} -1 & -2 & 1 & | & -2 \\ 3 & 4 & -2 & | & 0 \\ 2 & -4 & 0 & | & -4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & -1 & | & 2 \\ 3 & 4 & -2 & | & 0 \\ 2 & -4 & 0 & | & -4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & -1 & | & 2 \\ 0 & -2 & 1 & | & -6 \\ 2 & -4 & 0 & | & -4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & -1 & | & 2 \\ 0 & -2 & 1 & | & -6 \\ 0 & -8 & 2 & | & -8 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & -1 & | & 2 \\ 0 & 1 & -\frac{1}{2} & | & 3 \\ 0 & -8 & 2 & | & -8 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & | & -4 \\ 0 & 1 & -\frac{1}{2} & | & 3 \\ 0 & -8 & 2 & | & -8 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & | & -4 \\ 0 & 1 & -\frac{1}{2} & | & 3 \\ 0 & 0 & -2 & | & 16 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & | & -4 \\ 0 & 1 & -\frac{1}{2} & | & 3 \\ 0 & 0 & 1 & | & -8 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & | & -4 \\ 0 & 1 & 0 & | & -1 \\ 0 & 0 & 1 & | & -8 \end{bmatrix}$$

So $\vec{x} = (-2,0,-4)$, expressed in the alternate basis, is

$$[\vec{x}]_B = \begin{bmatrix} -4 \\ -1 \\ -8 \end{bmatrix}$$

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TRANSFORMATION MATRIX FOR A BASIS

1. Use the transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ to transform $[\vec{x}]_B = (2,1)$ in the basis B in the domain to a vector in the basis B in the codomain.

$$T(\vec{x}) = \begin{bmatrix} 3 & -2 \\ 6 & 0 \end{bmatrix} \vec{x}$$

$$B = \text{Span}\left(\begin{bmatrix} -2 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ -6 \end{bmatrix} \right)$$

Solution:

In order to transform a vector in the alternate basis in the domain into a vector in the alternate basis in the codomain, we need to find the transformation matrix M.

$$[T(\vec{x})]_B = M[\vec{x}]_B$$

We know that $M = C^{-1}AC$, and A was given to us in the problem as part of $T(\vec{x})$, so we just need to find C and C^{-1}.

The change of basis matrix C for the basis B is made of the column vectors that span B, $\vec{v}_1 = (-2,1)$ and $\vec{v}_2 = (4, -6)$, so

$$C = \begin{bmatrix} -2 & 4 \\ 1 & -6 \end{bmatrix}$$

Now we'll find C^{-1}.

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$$[C \mid I] = \begin{bmatrix} -2 & 4 & | & 1 & 0 \\ 1 & -6 & | & 0 & 1 \end{bmatrix}$$

$$[C \mid I] = \begin{bmatrix} 1 & -2 & | & -\frac{1}{2} & 0 \\ 1 & -6 & | & 0 & 1 \end{bmatrix}$$

$$[C \mid I] = \begin{bmatrix} 1 & -2 & | & -\frac{1}{2} & 0 \\ 0 & -4 & | & \frac{1}{2} & 1 \end{bmatrix}$$

$$[C \mid I] = \begin{bmatrix} 1 & -2 & | & -\frac{1}{2} & 0 \\ 0 & 1 & | & -\frac{1}{8} & -\frac{1}{4} \end{bmatrix}$$

$$[C \mid I] = \begin{bmatrix} 1 & 0 & | & -\frac{3}{4} & -\frac{1}{2} \\ 0 & 1 & | & -\frac{1}{8} & -\frac{1}{4} \end{bmatrix}$$

So,

$$C^{-1} = \begin{bmatrix} -\frac{3}{4} & -\frac{1}{2} \\ -\frac{1}{8} & -\frac{1}{4} \end{bmatrix}$$

With A, C, and C^{-1}, we can find $M = C^{-1}AC$.

$$M = \begin{bmatrix} -\frac{3}{4} & -\frac{1}{2} \\ -\frac{1}{8} & -\frac{1}{4} \end{bmatrix} \begin{bmatrix} 3 & -2 \\ 6 & 0 \end{bmatrix} \begin{bmatrix} -2 & 4 \\ 1 & -6 \end{bmatrix}$$

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$$M = \begin{bmatrix} -\frac{3}{4} & -\frac{1}{2} \\ -\frac{1}{8} & -\frac{1}{4} \end{bmatrix} \begin{bmatrix} 3(-2)-2(1) & 3(4)-2(-6) \\ 6(-2)+0(1) & 6(4)+0(-6) \end{bmatrix}$$

$$M = \begin{bmatrix} -\frac{3}{4} & -\frac{1}{2} \\ -\frac{1}{8} & -\frac{1}{4} \end{bmatrix} \begin{bmatrix} -6-2 & 12+12 \\ -12+0 & 24+0 \end{bmatrix}$$

$$M = \begin{bmatrix} -\frac{3}{4} & -\frac{1}{2} \\ -\frac{1}{8} & -\frac{1}{4} \end{bmatrix} \begin{bmatrix} -8 & 24 \\ -12 & 24 \end{bmatrix}$$

$$M = \begin{bmatrix} -\frac{3}{4}(-8)-\frac{1}{2}(-12) & -\frac{3}{4}(24)-\frac{1}{2}(24) \\ -\frac{1}{8}(-8)-\frac{1}{4}(-12) & -\frac{1}{8}(24)-\frac{1}{4}(24) \end{bmatrix}$$

$$M = \begin{bmatrix} 6+6 & -18-12 \\ 1+3 & -3-6 \end{bmatrix}$$

$$M = \begin{bmatrix} 12 & -30 \\ 4 & -9 \end{bmatrix}$$

We've been asked to transform $[\vec{x}]_B = (2,1)$, so we'll multiply M by this vector.

$$[T(\vec{x})]_B = M[\vec{x}]_B$$

$$[T(\vec{x})]_B = \begin{bmatrix} 12 & -30 \\ 4 & -9 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$[T(\vec{x})]_B = \begin{bmatrix} 12(2)-30(1) \\ 4(2)-9(1) \end{bmatrix}$$

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$$[T(\vec{x})]_B = \begin{bmatrix} -6 \\ -1 \end{bmatrix}$$

■ 2. Use the transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ to transform $\vec{x} = (-2,4)$ in the standard basis in the domain to a vector in the basis B in the codomain.

$$T(\vec{x}) = \begin{bmatrix} -3 & 1 \\ 4 & 5 \end{bmatrix} \vec{x}$$

$$B = \text{Span}\left(\begin{bmatrix} -2 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 3 \end{bmatrix} \right)$$

Solution:

The change of basis matrix C for the basis B is made of the column vectors that span B, $\vec{v}_1 = (-2,1)$ and $\vec{v}_2 = (-1,3)$, so

$$C = \begin{bmatrix} -2 & -1 \\ 1 & 3 \end{bmatrix}$$

Now we'll find C^{-1}.

$$[C \mid I] = \left[\begin{array}{cc|cc} -2 & -1 & 1 & 0 \\ 1 & 3 & 0 & 1 \end{array}\right]$$

$$[C \mid I] = \left[\begin{array}{cc|cc} 1 & \frac{1}{2} & -\frac{1}{2} & 0 \\ 1 & 3 & 0 & 1 \end{array}\right]$$

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$$[C \mid I] = \begin{bmatrix} 1 & \frac{1}{2} & | & -\frac{1}{2} & 0 \\ 0 & \frac{5}{2} & | & \frac{1}{2} & 1 \end{bmatrix}$$

$$[C \mid I] = \begin{bmatrix} 1 & \frac{1}{2} & | & -\frac{1}{2} & 0 \\ 0 & 1 & | & \frac{1}{5} & \frac{2}{5} \end{bmatrix}$$

$$[C \mid I] = \begin{bmatrix} 1 & 0 & | & -\frac{3}{5} & -\frac{1}{5} \\ 0 & 1 & | & \frac{1}{5} & \frac{2}{5} \end{bmatrix}$$

So,

$$C^{-1} = \begin{bmatrix} -\frac{3}{5} & -\frac{1}{5} \\ \frac{1}{5} & \frac{2}{5} \end{bmatrix}$$

Find the transformation $T(\vec{x})$ where $T(\vec{x}) = A\vec{x}$.

$$T(\vec{x}) = \begin{bmatrix} -3 & 1 \\ 4 & 5 \end{bmatrix} \vec{x}$$

$$T(\vec{x}) = \begin{bmatrix} -3 & 1 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} -2 \\ 4 \end{bmatrix}$$

$$T(\vec{x}) = \begin{bmatrix} -3(-2) + 1(4) \\ 4(-2) + 5(4) \end{bmatrix}$$

$$T(\vec{x}) = \begin{bmatrix} 6 + 4 \\ -8 + 20 \end{bmatrix}$$

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$$T(\vec{x}) = \begin{bmatrix} 10 \\ 12 \end{bmatrix}$$

Then $[T(\vec{x})]_B = C^{-1}T(\vec{x})$.

$$[T(\vec{x})]_B = \begin{bmatrix} -\frac{3}{5} & -\frac{1}{5} \\ \frac{1}{5} & \frac{2}{5} \end{bmatrix} \begin{bmatrix} 10 \\ 12 \end{bmatrix}$$

$$[T(\vec{x})]_B = \begin{bmatrix} -\frac{3}{5}(10) - \frac{1}{5}(12) \\ \frac{1}{5}(10) + \frac{2}{5}(12) \end{bmatrix}$$

$$[T(\vec{x})]_B = \begin{bmatrix} -\frac{30}{5} - \frac{12}{5} \\ \frac{10}{5} + \frac{24}{5} \end{bmatrix}$$

$$[T(\vec{x})]_B = \begin{bmatrix} -\frac{42}{5} \\ \frac{34}{5} \end{bmatrix}$$

■ 3. Use the transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ to transform $[\vec{x}]_B = (-5,2)$ in the basis B in the domain to a vector in the basis B in the codomain.

$$T(\vec{x}) = \begin{bmatrix} -2 & 3 \\ 1 & 5 \end{bmatrix} \vec{x}$$

$$B = \text{Span}\left(\begin{bmatrix} 1 \\ -3 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \end{bmatrix} \right)$$

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Solution:

In order to transform a vector in the alternate basis in the domain into a vector in the alternate basis in the codomain, we need to find the transformation matrix M.

$$[T(\overrightarrow{x})]_B = M[\overrightarrow{x}]_B$$

We know that $M = C^{-1}AC$, and A was given to us in the problem as part of $T(\overrightarrow{x})$, so we just need to find C and C^{-1}.

The change of basis matrix C for the basis B is made of the column vectors that span B, $\overrightarrow{v}_1 = (1, -3)$ and $\overrightarrow{v}_2 = (2,4)$, so

$$C = \begin{bmatrix} 1 & 2 \\ -3 & 4 \end{bmatrix}$$

Now we'll find C^{-1}.

$$[C \mid I] = \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ -3 & 4 & 0 & 1 \end{array}\right]$$

$$[C \mid I] = \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 10 & 3 & 1 \end{array}\right]$$

$$[C \mid I] = \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & \frac{3}{10} & \frac{1}{10} \end{array}\right]$$

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$$[C \mid I] = \begin{bmatrix} 1 & 0 & \mid & \frac{2}{5} & -\frac{1}{5} \\ 0 & 1 & \mid & \frac{3}{10} & \frac{1}{10} \end{bmatrix}$$

So,

$$C^{-1} = \begin{bmatrix} \frac{2}{5} & -\frac{1}{5} \\ \frac{3}{10} & \frac{1}{10} \end{bmatrix}$$

With A, C, and C^{-1}, we can find $M = C^{-1}AC$.

$$M = \begin{bmatrix} \frac{2}{5} & -\frac{1}{5} \\ \frac{3}{10} & \frac{1}{10} \end{bmatrix} \begin{bmatrix} -2 & 3 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -3 & 4 \end{bmatrix}$$

$$M = \begin{bmatrix} \frac{2}{5} & -\frac{1}{5} \\ \frac{3}{10} & \frac{1}{10} \end{bmatrix} \begin{bmatrix} -2(1)+3(-3) & -2(2)+3(4) \\ 1(1)+5(-3) & 1(2)+5(4) \end{bmatrix}$$

$$M = \begin{bmatrix} \frac{2}{5} & -\frac{1}{5} \\ \frac{3}{10} & \frac{1}{10} \end{bmatrix} \begin{bmatrix} -2-9 & -4+12 \\ 1-15 & 2+20 \end{bmatrix}$$

$$M = \begin{bmatrix} \frac{2}{5} & -\frac{1}{5} \\ \frac{3}{10} & \frac{1}{10} \end{bmatrix} \begin{bmatrix} -11 & 8 \\ -14 & 22 \end{bmatrix}$$

$$M = \begin{bmatrix} \frac{2}{5}(-11)-\frac{1}{5}(-14) & \frac{2}{5}(8)-\frac{1}{5}(22) \\ \frac{3}{10}(-11)+\frac{1}{10}(-14) & \frac{3}{10}(8)+\frac{1}{10}(22) \end{bmatrix}$$

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$$M = \begin{bmatrix} -\frac{22}{5} + \frac{14}{5} & \frac{16}{5} - \frac{22}{5} \\ -\frac{33}{10} - \frac{14}{10} & \frac{24}{10} + \frac{22}{10} \end{bmatrix}$$

$$M = \begin{bmatrix} -\frac{8}{5} & -\frac{6}{5} \\ -\frac{47}{10} & \frac{46}{10} \end{bmatrix}$$

We've been asked to transform $[\vec{x}]_B = (-5,2)$, so we'll multiply M by this vector.

$$[T(\vec{x})]_B = M[\vec{x}]_B$$

$$[T(\vec{x})]_B = \begin{bmatrix} -\frac{8}{5} & -\frac{6}{5} \\ -\frac{47}{10} & \frac{46}{10} \end{bmatrix} \begin{bmatrix} -5 \\ 2 \end{bmatrix}$$

$$[T(\vec{x})]_B = \begin{bmatrix} -\frac{8}{5}(-5) - \frac{6}{5}(2) \\ -\frac{47}{10}(-5) + \frac{46}{10}(2) \end{bmatrix}$$

$$[T(\vec{x})]_B = \begin{bmatrix} \frac{40}{5} - \frac{12}{5} \\ \frac{235}{10} + \frac{92}{10} \end{bmatrix}$$

$$[T(\vec{x})]_B = \begin{bmatrix} \frac{28}{5} \\ \frac{327}{10} \end{bmatrix}$$

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4. Use the transformation $T : \mathbb{R}^3 \to \mathbb{R}^3$ to transform $\vec{x} = (6, -3)$ in the standard basis in the domain to a vector in the basis B in the codomain.

$$T(\vec{x}) = \begin{bmatrix} -5 & -4 \\ 2 & -8 \end{bmatrix} \vec{x}$$

$$B = \text{Span}\left(\begin{bmatrix} 1 \\ -2 \end{bmatrix}, \begin{bmatrix} 0 \\ 4 \end{bmatrix} \right)$$

Solution:

The change of basis matrix C for the basis B is made of the column vectors that span B, $\vec{v}_1 = (1, -2)$ and $\vec{v}_2 = (0,4)$, so

$$C = \begin{bmatrix} 1 & 0 \\ -2 & 4 \end{bmatrix}$$

Now we'll find C^{-1}.

$$[C \mid I] = \left[\begin{array}{cc|cc} 1 & 0 & 1 & 0 \\ -2 & 4 & 0 & 1 \end{array}\right]$$

$$[C \mid I] = \left[\begin{array}{cc|cc} 1 & 0 & 1 & 0 \\ 0 & 4 & 2 & 1 \end{array}\right]$$

$$[C \mid I] = \left[\begin{array}{cc|cc} 1 & 0 & 1 & 0 \\ 0 & 1 & \frac{1}{2} & \frac{1}{4} \end{array}\right]$$

So,

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$$C^{-1} = \begin{bmatrix} 1 & 0 \\ \frac{1}{2} & \frac{1}{4} \end{bmatrix}$$

Find the transformation $T(\vec{x})$ where $T(\vec{x}) = A\vec{x}$.

$$T(\vec{x}) = \begin{bmatrix} -5 & -4 \\ 2 & -8 \end{bmatrix} \vec{x}$$

$$T(\vec{x}) = \begin{bmatrix} -5 & -4 \\ 2 & -8 \end{bmatrix} \begin{bmatrix} 6 \\ -3 \end{bmatrix}$$

$$T(\vec{x}) = \begin{bmatrix} -5(6) - 4(-3) \\ 2(6) - 8(-3) \end{bmatrix}$$

$$T(\vec{x}) = \begin{bmatrix} -30 + 12 \\ 12 + 24 \end{bmatrix}$$

$$T(\vec{x}) = \begin{bmatrix} -18 \\ 36 \end{bmatrix}$$

Then $[T(\vec{x})]_B = C^{-1}T(\vec{x})$.

$$[T(\vec{x})]_B = \begin{bmatrix} 1 & 0 \\ \frac{1}{2} & \frac{1}{4} \end{bmatrix} \begin{bmatrix} -18 \\ 36 \end{bmatrix}$$

$$[T(\vec{x})]_B = \begin{bmatrix} 1(-18) + 0(36) \\ \frac{1}{2}(-18) + \frac{1}{4}(36) \end{bmatrix}$$

$$[T(\vec{x})]_B = \begin{bmatrix} -18 \\ 0 \end{bmatrix}$$

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■ 5. Use the transformation $T : \mathbb{R}^3 \to \mathbb{R}^3$ to transform $[\vec{x}]_B = (-2,4,1)$ in the basis B in the domain to a vector in the basis B in the codomain.

$$T(\vec{x}) = \begin{bmatrix} -4 & 1 & 1 \\ 2 & -3 & -1 \\ 0 & 2 & 0 \end{bmatrix} \vec{x}$$

$$B = \text{Span}\left(\begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right)$$

Solution:

In order to transform a vector in the alternate basis in the domain into a vector in the alternate basis in the codomain, we need to find the transformation matrix M.

$$[T(\vec{x})]_B = M[\vec{x}]_B$$

We know that $M = C^{-1}AC$, and A was given to us in the problem as part of $T(\vec{x})$, so we just need to find C and C^{-1}.

The change of basis matrix C that transforms vectors from the standard basis into vectors in the alternate basis B is made of the column vectors that span B, $\vec{v}_1 = (1, -1, 2)$, $\vec{v}_2 = (0, 2, -1)$, and $\vec{v}_3 = (1, 0, 1)$, so

$$C = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 2 & 0 \\ 2 & -1 & 1 \end{bmatrix}$$

Now we'll find C^{-1}.

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$$[C \mid I] = \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ -1 & 2 & 0 & 0 & 1 & 0 \\ 2 & -1 & 1 & 0 & 0 & 1 \end{array}\right]$$

$$[C \mid I] = \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 2 & 1 & 1 & 1 & 0 \\ 2 & -1 & 1 & 0 & 0 & 1 \end{array}\right]$$

$$[C \mid I] = \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 2 & 1 & 1 & 1 & 0 \\ 0 & -1 & -1 & -2 & 0 & 1 \end{array}\right]$$

$$[C \mid I] = \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & -1 & -1 & -2 & 0 & 1 \end{array}\right]$$

$$[C \mid I] = \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & -\frac{1}{2} & -\frac{3}{2} & \frac{1}{2} & 1 \end{array}\right]$$

$$[C \mid I] = \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 & 3 & -1 & -2 \end{array}\right]$$

$$[C \mid I] = \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -2 & 1 & 2 \\ 0 & 1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 & 3 & -1 & -2 \end{array}\right]$$

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$$[C \mid I] = \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -2 & 1 & 2 \\ 0 & 1 & 0 & -1 & 1 & 1 \\ 0 & 0 & 1 & 3 & -1 & -2 \end{array}\right]$$

So,

$$C^{-1} = \begin{bmatrix} -2 & 1 & 2 \\ -1 & 1 & 1 \\ 3 & -1 & -2 \end{bmatrix}$$

With A, C, and C^{-1}, we can find $M = C^{-1}AC$.

$$M = \begin{bmatrix} -2 & 1 & 2 \\ -1 & 1 & 1 \\ 3 & -1 & -2 \end{bmatrix} \begin{bmatrix} -4 & 1 & 1 \\ 2 & -3 & -1 \\ 0 & 2 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ -1 & 2 & 0 \\ 2 & -1 & 1 \end{bmatrix}$$

$$M = \begin{bmatrix} -2 & 1 & 2 \\ -1 & 1 & 1 \\ 3 & -1 & -2 \end{bmatrix} \begin{bmatrix} -4(1)+1(-1)+1(2) & -4(0)+1(2)+1(-1) & -4(1)+1(0)+1(1) \\ 2(1)-3(-1)-1(2) & 2(0)-3(2)-1(-1) & 2(1)-3(0)-1(1) \\ 0(1)+2(-1)+0(2) & 0(0)+2(2)+0(-1) & 0(1)+2(0)+0(1) \end{bmatrix}$$

$$M = \begin{bmatrix} -2 & 1 & 2 \\ -1 & 1 & 1 \\ 3 & -1 & -2 \end{bmatrix} \begin{bmatrix} -4-1+2 & 0+2-1 & -4+0+1 \\ 2+3-2 & 0-6+1 & 2-0-1 \\ 0-2+0 & 0+4+0 & 0+0+0 \end{bmatrix}$$

$$M = \begin{bmatrix} -2 & 1 & 2 \\ -1 & 1 & 1 \\ 3 & -1 & -2 \end{bmatrix} \begin{bmatrix} -3 & 1 & -3 \\ 3 & -5 & 1 \\ -2 & 4 & 0 \end{bmatrix}$$

$$M = \begin{bmatrix} -2(-3)+1(3)+2(-2) & -2(1)+1(-5)+2(4) & -2(-3)+1(1)+2(0) \\ -1(-3)+1(3)+1(-2) & -1(1)+1(-5)+1(4) & -1(-3)+1(1)+1(0) \\ 3(-3)-1(3)-2(-2) & 3(1)-1(-5)-2(4) & 3(-3)-1(1)-2(0) \end{bmatrix}$$

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$$M = \begin{bmatrix} 6+3-4 & -2-5+8 & 6+1+0 \\ 3+3-2 & -1-5+4 & 3+1+0 \\ -9-3+4 & 3+5-8 & -9-1-0 \end{bmatrix}$$

$$M = \begin{bmatrix} 5 & 1 & 7 \\ 4 & -2 & 4 \\ -8 & 0 & -10 \end{bmatrix}$$

We've been asked to transform $[\vec{x}]_B = (-2,4,1)$, so we'll multiply M by this vector.

$$[T(\vec{x})]_B = M[\vec{x}]_B$$

$$[T(\vec{x})]_B = \begin{bmatrix} 5 & 1 & 7 \\ 4 & -2 & 4 \\ -8 & 0 & -10 \end{bmatrix} \begin{bmatrix} -2 \\ 4 \\ 1 \end{bmatrix}$$

$$[T(\vec{x})]_B = \begin{bmatrix} 5(-2)+1(4)+7(1) \\ 4(-2)-2(4)+4(1) \\ -8(-2)+0(4)-10(1) \end{bmatrix}$$

$$[T(\vec{x})]_B = \begin{bmatrix} -10+4+7 \\ -8-8+4 \\ 16+0-10 \end{bmatrix}$$

$$[T(\vec{x})]_B = \begin{bmatrix} 1 \\ -12 \\ 6 \end{bmatrix}$$

■ 6. Use the transformation $T : \mathbb{R}^3 \to \mathbb{R}^3$ to transform $\vec{x} = (-2,3,1)$ in the standard basis in the domain to a vector in the basis B in the codomain.

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$$T(\vec{x}) = \begin{bmatrix} -4 & 2 & 1 \\ 0 & 3 & -5 \\ 1 & -2 & 4 \end{bmatrix} \vec{x}$$

$$B = \text{Span}\left(\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ -1 \end{bmatrix} \right)$$

Solution:

The change of basis matrix C for the basis B is made of the column vectors that span B, $\vec{v}_1 = (-1,1,0)$, $\vec{v}_2 = (-2,1,-1)$, and $\vec{v}_3 = (-2,0,-1)$, so

$$C = \begin{bmatrix} -1 & -2 & -2 \\ 1 & 1 & 0 \\ 0 & -1 & -1 \end{bmatrix}$$

Now we'll find C^{-1}.

$$[C \mid I] = \left[\begin{array}{ccc|ccc} -1 & -2 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & -1 & -1 & 0 & 0 & 1 \end{array}\right]$$

$$[C \mid I] = \left[\begin{array}{ccc|ccc} 1 & 2 & 2 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & -1 & -1 & 0 & 0 & 1 \end{array}\right]$$

$$[C \mid I] = \left[\begin{array}{ccc|ccc} 1 & 2 & 2 & -1 & 0 & 0 \\ 0 & -1 & -2 & 1 & 1 & 0 \\ 0 & -1 & -1 & 0 & 0 & 1 \end{array}\right]$$

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$$[C \mid I] = \begin{bmatrix} 1 & 2 & 2 & | & -1 & 0 & 0 \\ 0 & 1 & 2 & | & -1 & -1 & 0 \\ 0 & -1 & -1 & | & 0 & 0 & 1 \end{bmatrix}$$

$$[C \mid I] = \begin{bmatrix} 1 & 0 & -2 & | & 1 & 2 & 0 \\ 0 & 1 & 2 & | & -1 & -1 & 0 \\ 0 & -1 & -1 & | & 0 & 0 & 1 \end{bmatrix}$$

$$[C \mid I] = \begin{bmatrix} 1 & 0 & -2 & | & 1 & 2 & 0 \\ 0 & 1 & 2 & | & -1 & -1 & 0 \\ 0 & 0 & 1 & | & -1 & -1 & 1 \end{bmatrix}$$

$$[C \mid I] = \begin{bmatrix} 1 & 0 & 0 & | & -1 & 0 & 2 \\ 0 & 1 & 2 & | & -1 & -1 & 0 \\ 0 & 0 & 1 & | & -1 & -1 & 1 \end{bmatrix}$$

$$[C \mid I] = \begin{bmatrix} 1 & 0 & 0 & | & -1 & 0 & 2 \\ 0 & 1 & 0 & | & 1 & 1 & -2 \\ 0 & 0 & 1 & | & -1 & -1 & 1 \end{bmatrix}$$

So,

$$C^{-1} = \begin{bmatrix} -1 & 0 & 2 \\ 1 & 1 & -2 \\ -1 & -1 & 1 \end{bmatrix}$$

Find the transformation $T(\vec{x})$ where $T(\vec{x}) = A\vec{x}$.

$$T(\vec{x}) = \begin{bmatrix} -4 & 2 & 1 \\ 0 & 3 & -5 \\ 1 & -2 & 4 \end{bmatrix} \vec{x}$$

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$$T(\vec{x}) = \begin{bmatrix} -4 & 2 & 1 \\ 0 & 3 & -5 \\ 1 & -2 & 4 \end{bmatrix} \begin{bmatrix} -2 \\ 3 \\ 1 \end{bmatrix}$$

$$T(\vec{x}) = \begin{bmatrix} -4(-2) + 2(3) + 1(1) \\ 0(-2) + 3(3) - 5(1) \\ 1(-2) - 2(3) + 4(1) \end{bmatrix}$$

$$T(\vec{x}) = \begin{bmatrix} 8 + 6 + 1 \\ 0 + 9 - 5 \\ -2 - 6 + 4 \end{bmatrix}$$

$$T(\vec{x}) = \begin{bmatrix} 15 \\ 4 \\ -4 \end{bmatrix}$$

Then $[T(\vec{x})]_B = C^{-1}T(\vec{x})$.

$$[T(\vec{x})]_B = \begin{bmatrix} -1 & 0 & 2 \\ 1 & 1 & -2 \\ -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 15 \\ 4 \\ -4 \end{bmatrix}$$

$$[T(\vec{x})]_B = \begin{bmatrix} -1(15) + 0(4) + 2(-4) \\ 1(15) + 1(4) - 2(-4) \\ -1(15) - 1(4) + 1(-4) \end{bmatrix}$$

$$[T(\vec{x})]_B = \begin{bmatrix} -15 + 0 - 8 \\ 15 + 4 + 8 \\ -15 - 4 - 4 \end{bmatrix}$$

$$[T(\vec{x})]_B = \begin{bmatrix} -23 \\ 27 \\ -23 \end{bmatrix}$$

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